VICREG: VARIANCE-INVARIANCE-COVARIANCE REGULARIZATION FOR SELF-SUPERVISED LEARNING

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ABSTRACT

Recent self-supervised methods for image representation learning maximize the agreement between embedding vectors produced by encoders fed with different views of the same image. The main challenge is to prevent a *collapse* in which the encoders produce constant or non-informative vectors. We introduce VICReg (Variance-Invariance-Covariance Regularization), a method that explicitly avoids the collapse problem with two regularizations terms applied to both embeddings separately: (1) a term that maintains the variance of each embedding dimension above a threshold, (2) a term that decorrelates each pair of variables. Unlike most other approaches to the same problem, VICReg does *not* require techniques such as: weight sharing between the branches, batch normalization, feature-wise normalization, output quantization, stop gradient, memory banks, etc., and achieves results on par with the state of the art on several downstream tasks. In addition, we show that our variance regularization term stabilizes the training of other methods and leads to performance improvements.

1 INTRODUCTION

Self-supervised representation learning has made significant progress over the last years, almost reaching the performance of supervised baselines on many downstream tasks Bachman et al. (2019); Misra & Maaten (2020); He et al. (2020); Tian et al. (2020); Caron et al. (2020); Grill et al. (2020); Chen & He (2020); Gidaris et al. (2021); Zbontar et al. (2021). Several recent approaches rely on a *joint embedding architecture* in which two networks are trained to produce similar embeddings for different views of the same image. A popular instance is the Siamese network architecture Bromley et al. (1994), where the two networks share the same weights. The main challenge with joint embedding architectures is to prevent a *collapse* in which the two branches ignore the inputs and produce identical and constant output vectors. There are two main approaches to preventing collapse: contrastive methods and information maximization methods. Contrastive Bromley et al. (1994); Chopra et al. (2005); He et al. (2020); Hjelm et al. (2019); Chen et al. (2020a) methods tend to be costly, require large batch sizes or memory banks, and use a loss that explicitly pushes the embeddings of dissimilar images away from each other. They often require a mining procedure to search for offending dissimilar samples from a memory bank He et al. (2020) or from the current batch Chen et al. (2020a). Quantization-based approaches Caron et al. (2020; 2018) force the embeddings of different samples to belong to different clusters on the unit sphere. Collapse is prevented by ensuring that the assignment of samples to clusters is as uniform as possible. A similarity term encourages the cluster assignment score vectors from the two branches to be similar. More recently, a few methods have appeared that do not rely on contrastive samples or vector quantization, yet produce high-quality representations, for example BYOL Grill et al. (2020) and SimSiam Chen & He (2020). They exploit several tricks: batch-wise or feature-wise normalization, a "momentum encoder" in which the parameter vector of one branch is a low-pass-filtered version of the parameter vector of the other branch Grill et al. (2020); Richemond et al. (2020), or a stop-gradient operation in one of the branches Chen & He (2020). The dynamics of learning in these methods, and how they avoid collapse, is not fully understood, although theoretical and empirical studies point to the crucial importance of batch-wise or feature-wise normalization Richemond et al. (2020); Tian et al. (2021). Finally, an

VICREG：用于自监督学习的方差—不变性—协方差正则化

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摘要

近期图像表征自监督方法最大化编码器对同一图像不同视图所生成嵌入向量的一致性。主要挑战是防止坍塌，即编码器输出常量或不含有效信息的向量。我们提出 VICReg（方差—不变性—协方差正则化），通过分别施加于两组嵌入的两个正则项显式避免坍塌：（1）让每个嵌入维度的方差保持在阈值以上；（2）让每对变量去相关。与多数解决这一问题的方法不同，VICReg 不需要分支间权重共享、批归一化、按特征归一化、输出量化、停止梯度、记忆库等技术，却在多个下游任务中取得与最优方法相当的结果。此外，方差正则项能稳定其他方法的训练，并提高性能。

1. 引言

近几年，自监督表征学习进展显著，在许多下游任务中几乎达到监督基线性能（Bachman et al., 2019; Misra & Maaten, 2020; He et al., 2020; Tian et al., 2020; Caron et al., 2020; Grill et al., 2020; Chen & He, 2020; Gidaris et al., 2021; Zbontar et al., 2021）。近期多种方法依赖联合嵌入结构，训练两个网络为同一图像不同视图产生相似嵌入。常见实例是孪生网络（Bromley et al., 1994），两个网络共享权重。主要挑战是防止坍塌：两分支忽略输入，输出相同的常量向量。防止坍塌主要有两条路线：对比方法和信息最大化方法。对比方法（Bromley et al., 1994; Chopra et al., 2005; He et al., 2020; Hjelm et al., 2019; Chen et al., 2020a）成本往往较高，需要大批量或记忆库，损失显式推远不同图像的嵌入。通常还要从记忆库（He et al., 2020）或当前批次（Chen et al., 2020a）中挖掘容易混淆的不同样本。基于量化的方法（Caron et al., 2020; 2018）迫使不同样本的嵌入属于单位球面上的不同聚类，通过尽量均匀分配样本防坍塌，并用相似性项让两分支聚类分配得分向量相似。更近期的 BYOL（Grill et al., 2020）、SimSiam（Chen & He, 2020）等无需对比样本或向量量化，也能产生优质表征。它们使用多种技巧：沿批量或特征归一化；“动量编码器”，即一侧参数向量是另一侧的低通滤波版本（Grill et al., 2020; Richemond et al., 2020）；或一侧停止梯度（Chen & He, 2020）。这些方法的学习动态及防坍塌原理尚未完全清楚，尽管理论与实验提示沿批量或特征归一化至关重要（Richemond et al., 2020; Tian et al., 2021）。最后，一种

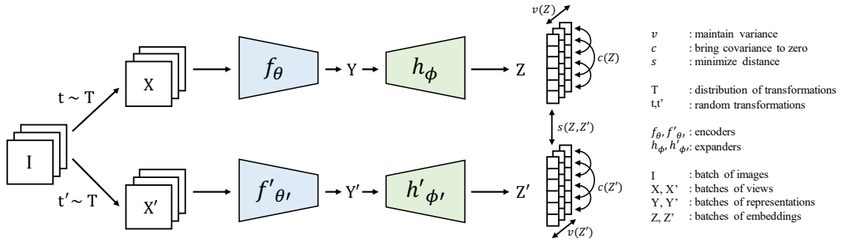


Figure 1: **VICReg: joint embedding architecture with variance, invariance and covariance regularization.** Given a batch of images I, two batches of different views X and X' are produced and are then encoded into representations Y and Y'. The representations are fed to an expander producing the embeddings Z and Z'. The distance between two embeddings from the same image is minimized, the variance of each embedding variable over a batch is maintained above a threshold, and the covariance between pairs of embedding variables over a batch are attracted to zero, decorrelating the variables from each other. Although the two branches do not require identical architectures nor share weights, in most of our experiments, they are Siamese with shared weights: the encoders are ResNet-50 backbones with output dimension 2048. The expanders have 3 fully-connected layers of size 8192.

alternative class of collapse prevention methods relies on maximizing the information content of the embedding Zbontar et al. (2021); Ermolov et al. (2021). These methods prevent *informational collapse* by decorrelating every pair of variables of the embedding vectors. This indirectly maximizes the information content of the embedding vectors. The Barlow Twins method drives the normalized cross-correlation matrix of the two embeddings towards the identity Zbontar et al. (2021), while the Whitening-MSE method whitens and spreads out the embedding vectors on the unit sphere Ermolov et al. (2021).

2 VICREG: INTUITION

We introduce VICReg (Variance-Invariance-Covariance Regularization), a self-supervised method for training joint embedding architectures based on the principle of preserving the information content of the embeddings. The basic idea is to use a loss function with three terms:

- **Invariance**: the mean square distance between the embedding vectors.
- **Variance**: a hinge loss to maintain the standard deviation (over a batch) of each variable of the embedding above a given threshold. This term forces the embedding vectors of samples within a batch to be different.
- **Covariance**: a term that attracts the covariances (over a batch) between every pair of (centered) embedding variables towards zero. This term decorrelates the variables of each embedding and prevents an *informational collapse* in which the variables would vary together or be highly correlated.

Variance and Covariance terms are applied to both branches of the architecture separately, thereby preserving the information content of each embedding at a certain level and preventing informational collapse independently for the two branches. The main contribution of this paper is the Variance preservation term, which explicitly prevents a collapse due to a shrinkage of the embedding vectors towards zero. The Covariance criterion is borrowed from the Barlow Twins method and prevents informational collapse due to redundancy between the embedding variables Zbontar et al. (2021). VICReg is more generally applicable than most of the aforementioned methods because of fewer constraints on the architecture. In particular, VICReg:

- does not require that the weights of the two branches be shared, not that the architectures be identical, nor that the inputs be of the same nature;

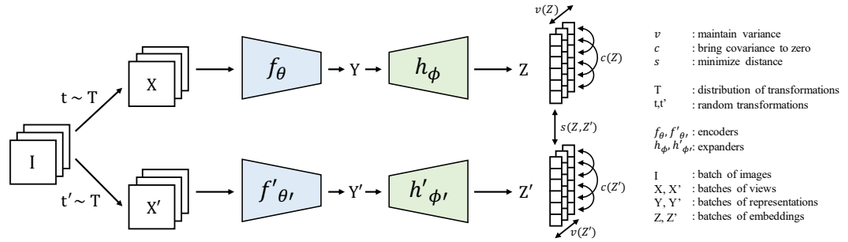


图1：VICReg：采用方差、不变性和协方差正则化的联合嵌入结构。给定一批图像 I，产生两批不同视图 X、X'，编码成表征 Y、Y'，再由扩展器生成嵌入 Z、Z'。最小化同一图像的两种嵌入距离，令每个嵌入变量在批内的方差保持在阈值以上，并使变量对的批内协方差趋向零，以去相关。虽然两分支无需相同结构或共享权重，多数实验仍采用共享权重孪生网络：编码器为2048维 ResNet-50；扩展器为三层8192维全连接层。

另一类防坍塌方法通过最大化嵌入的信息量实现（Zbontar et al., 2021; Ermolov et al., 2021）。它们使嵌入向量每对变量去相关，以防信息坍塌，并间接最大化嵌入信息量。Barlow Twins 将两组嵌入的归一化交叉相关矩阵推向单位矩阵（Zbontar et al., 2021）；Whitening-MSE 将嵌入白化，并使其在单位球面分散（Ermolov et al., 2021）。

2. VICREG：直觉

我们提出 VICReg（方差—不变性—协方差正则化），基于保持嵌入信息量的原则，训练自监督联合嵌入结构。基本思路是使用三项损失：

- 不变性：嵌入向量间的平均平方距离。
- 方差：铰链损失，保持每个嵌入变量的批内标准差高于给定阈值，迫使同批样本嵌入向量彼此不同。
- 协方差：使每对中心化嵌入变量的批内协方差趋向零。该项让每种嵌入的变量去相关，防止变量共同变化或高度相关的信息坍塌。

方差与协方差项分别用于两个分支，让各嵌入的信息量保持在一定水平，并独立防止信息坍塌。本文主要贡献是方差保持项，显式防止嵌入向量缩至零而坍塌。协方差准则借鉴 Barlow Twins，防止嵌入变量冗余引起的信息坍塌（Zbontar et al., 2021）。结构约束较少，使 VICReg 比上述多数方法适用范围更广。具体而言，VICReg：

- 无需两分支共享权重、采用相同结构，也无需输入性质相同；

- does not require a memory bank, nor contrastive samples, nor a large batch size;
- does not require batch-wise nor feature-wise normalization; and
- does not require vector quantization nor a predictor module.

Other methods require asymmetric stop gradient operations, as in SimSiam Chen & He (2020), weight sharing between the two branches as in classical Siamese nets, or weight sharing through exponential moving average dampening with stop gradient in one branch, as in BYOL and MoCo He et al. (2020); Grill et al. (2020); Chen et al. (2020c), large batches of contrastive samples, as in SimCLR Chen et al. (2020a), or batch-wise and/or feature-wise normalization Caron et al. (2020); Grill et al. (2020); Chen & He (2020); Zbontar et al. (2021); Ermolov et al. (2021). One of the most interesting feature of VICReg is the fact that the two branches are not required to share the same parameters, architecture, or input modality. This opens the door to the use of non-contrastive self-supervised joint-embedding for multi-modal signals, such as video and audio. We demonstrate the effectiveness of the proposed approach by evaluating the representations learned with VICReg on several downstream image recognition tasks including linear head and semi-supervised evaluation protocols for image classification on ImageNet Deng et al. (2009), and other classification, detection, instance segmentation, and retrieval tasks. Furthermore, we show that incorporating variance preservation into other self-supervised joint-embedding methods yields better training stability and performance improvement on downstream tasks. More generally, we show that VICReg is an explicit and effective, yet simple method for preventing collapse in self-supervised joint-embedding learning.

3 Related work

Contrastive learning. In contrastive SSL methods applied to joint embedding architectures, the output embeddings for a sample and its distorted version are brought close to each other, while other samples and their distortions are pushed away. The method is most often applied to Siamese architectures in which the two branches have identical architectures and share weights Misra & Maaten (2020); He et al. (2020); Bromley et al. (1994); Hjelm et al. (2019); Chen et al. (2020a;c); Hadsell et al. (2006); Ye et al. (2019); Wu et al. (2018); van den Oord et al. (2018); Chen et al. (2020b). Many authors use the InfoNCE loss van den Oord et al. (2018) in which the repulsive force is larger for contrastive samples that are closer to the reference. While these methods yield good performance, they require large amounts of contrastive pairs in order to work well. These contrastive pairs can be sampled from a memory bank as in MoCo He et al. (2020), or given by the current batch of data as in SimCLR Chen et al. (2020a), with a significant memory footprint. This downside of contrastive methods motivates a search for alternatives.

Clustering methods. Instead of viewing each sample as its own class, clustering-based methods group them into clusters based on some similarity measure Caron et al. (2020; 2018); Bautista et al. (2016); Yang et al. (2016); Xie et al. (2016); Huang et al. (2019); Zhuang et al. (2019); Caron et al. (2019); Asano et al. (2020); Yan et al. (2020). DeepCluster Caron et al. (2018) uses k-means assignments of representations from previous iterations as pseudo-labels for the new representations, which requires an expensive clustering phase done asynchronously, and makes the method hard to scale up. SwAV Caron et al. (2020) mitigates this issue by learning the clusters online while maintaining a balanced partition of the assignments through the Sinkhorn-Knopp transform Cuturi (2013). These clustering approaches can be viewed as contrastive learning at the level of clusters which still requires a lot of negative comparisons to work well.

Distillation methods. Recent proposals such as BYOL, SimSiam, OBoW and variants Grill et al. (2020); Chen & He (2020); Gidaris et al. (2021); Richemond et al. (2020); Gidaris et al. (2020) have shown that collapse can be avoided by using architectural tricks inspired by knowledge distillation Hinton et al. (2015). These methods train a student network to predict the representations of a teacher network, for which the weights are a running average of the student network's weights Grill et al. (2020), or are shared with the student network, but no gradient is back-propagated through the teacher Chen & He (2020). These methods are effective, but there is no clear understanding of why and how they avoid collapse. Alternatively, the images can be represented as bags of word over a dictionary of visual features, which effectively prevents collapse. In OBoW Gidaris et al. (2020) and Gidaris et al. (2021) the dictionary is obtained by off-line or on-line clustering. By contrast, our method explicitly prevents collapse in the two branches independently, which removes

- 无需记忆库、对比样本或大批量；
- 无需沿批量或特征维度归一化；并且
- 无需向量量化或预测器模块。

其他方法需要非对称停止梯度，如 SimSiam（Chen & He, 2020）；需要两个分支共享权重，如经典孪生网络；或通过指数移动平均平滑共享权重，并在一侧停止梯度，如 BYOL 和 MoCo（He et al., 2020; Grill et al., 2020; Chen et al., 2020c）；需要大批量对比样本，如 SimCLR（Chen et al., 2020a）；或需要沿批量及／或特征维度归一化（Caron et al., 2020; Grill et al., 2020; Chen & He, 2020; Zbontar et al., 2021; Ermolov et al., 2021）。VICReg 最有趣的特性之一，是两个分支无需共享参数、结构或输入模态。这为视频与音频等多模态信号的非对比式自监督联合嵌入打开了道路。我们在多个下游图像识别任务上验证其有效性，包括 ImageNet（Deng et al., 2009）图像分类的线性分类头及半监督评估，以及其他分类、检测、实例分割和检索任务。进一步表明，在其他自监督联合嵌入方法中加入方差保持，能改善训练稳定性和下游性能。总体而言，VICReg 是一种显式、有效且简单的联合嵌入自监督防坍塌方法。

3. 相关工作

对比学习 联合嵌入结构的对比式 SSL，将样本及其扰动版本的嵌入拉近，将其他样本及其扰动推远。通常使用两分支结构相同、共享权重的孪生网络（Misra & Maaten, 2020; He et al., 2020; Bromley et al., 1994; Hjelm et al., 2019; Chen et al., 2020a;c; Hadsell et al., 2006; Ye et al., 2019; Wu et al., 2018; van den Oord et al., 2018; Chen et al., 2020b）。许多工作采用 InfoNCE（van den Oord et al., 2018），越靠近参考样本的对比样本，受到的排斥越强。这些方法虽有效，但需要大量对比样本对。可从 MoCo 的记忆库（He et al., 2020）抽取，或由 SimCLR 当前批次（Chen et al., 2020a）提供，内存占用都较高。因此，需要探索替代方案。

聚类方法 聚类方法不把每个样本视为独立类别，而是按相似性分组（Caron et al., 2020; 2018; Bautista et al., 2016; Yang et al., 2016; Xie et al., 2016; Huang et al., 2019; Zhuang et al., 2019; Caron et al., 2019; Asano et al., 2020; Yan et al., 2020）。DeepCluster（Caron et al., 2018）以前次迭代表征的 k-means 分配为新表征的伪标签，需要昂贵的异步聚类阶段，难以扩展。SwAV（Caron et al., 2020）在线学习聚类，同时用 Sinkhorn-Knopp 变换（Cuturi, 2013）维持均衡分配，缓解这一问题。这些方法可视为聚类层面的对比学习，仍需大量负向比较。

蒸馏方法 BYOL、SimSiam、OBoW 及其变体等近期方法（Grill et al., 2020; Chen & He, 2020; Gidaris et al., 2021; Richemond et al., 2020; Gidaris et al., 2020），通过受知识蒸馏（Hinton et al., 2015）启发的结构技巧避免坍塌。学生网络预测教师表征；教师权重是学生权重的移动平均（Grill et al., 2020），或与学生共享，但不通过教师反向传播梯度（Chen & He, 2020）。这些方法有效，却缺乏对其防坍塌原理的清晰理解。也可将图像表示为视觉特征字典上的词袋，以有效防坍塌。OBoW（Gidaris et al., 2020; 2021）通过离线或在线聚类构建字典。相比之下，本方法在两分支各自显式防坍塌，消除了

the requirement for shared weights and identical architecture, opening the door to the application of joint-embedding SSL to multi-modal signals.

Information maximization methods. A principle to prevent collapse is to maximize the information content of the embeddings. Two such methods were recently proposed: W-MSE Ermolov et al. (2021) and Barlow Twins Zbontar et al. (2021). In W-MSE, an extra module transforms the embeddings into the eigenspace of their covariance matrix (whitening or Karhunen-Loève transform), and forces the vectors thereby obtained to be uniformly distributed on the unit sphere. In Barlow Twins, a loss term attempts to make the normalized cross-correlation matrix of the embedding vectors from the two branches to be close to the identity. Both methods attempt to produce embedding variables that are decorrelated from each other, thus preventing an *informational collapse* in which the variables carry redundant information. Because all variables are normalized over a batch, there is no incentive for them to shrink nor expand. This seems to sufficient to prevent collapse. Our method borrows the decorrelation mechanism of Barlow Twins. But it includes an explicit variance-preservation term for each variable of the two embeddings and thus does not require any normalization.

4 VICREG: DETAILED DESCRIPTION

VICReg follows recent trends in self-supervised learning Caron et al. (2020); Grill et al. (2020); Chen & He (2020); Zbontar et al. (2021); Chen et al. (2020a) and is based on a *joint embedding architecture*. Contrary to many previous approaches, our architecture may be completely symmetric or completely asymmetric with no shared structure or parameters between the two branches. In most of our experiments, we use a Siamese net architecture in which the two branches are identical and share weights. Each branch consists of an *encoder* f_θ that outputs the representations (used for downstream tasks), followed by an *expander* h_ϕ that maps the representations into an embedding space where the loss function will be computed. The role of the expander is twofold: (1) eliminate the information by which the two representations differ, (2) expand the dimension in a non-linear fashion so that decorrelating the embedding variables will reduce the dependencies (not just the correlations) between the variables of the representation vector. The loss function uses a term s that learns invariance to data transformations and is regularized with a variance term v that prevents norm collapse and a covariance term c that prevents informational collapse by decorrelating the different dimensions of the vectors. After pretraining, the expander is discarded and the representations of the encoder are used for downstream tasks.

4.1 METHOD

Given an image i sampled from a dataset $\mathcal{D}$, two transformations t and t' are sampled from a distribution $\mathcal{T}$ to produce two different views $x = t(i)$ and $x' = t'(i)$ of i. These transformations are random crops of the image, followed by color distortions. The distribution $\mathcal{T}$ is described in Appendix C. The views x and x' are first encoded by f_θ into their *representations* $y = f_\theta(x)$ and $y' = f_\theta(x')$, which are then mapped by the expander h_ϕ onto the *embeddings* $z = h_\phi(y)$ and $z' = h_\phi(y')$. The loss is computed at the embedding level on z and z'.

We describe here the variance, invariance and covariance terms that compose our loss function. The images are processed in batches, and we denote $Z = [z_1, \dots, z_n]$ and $Z' = [z'_1, \dots, z'_n]$ the two batches composed of n vectors of dimension d, of embeddings coming out of the two branches of the siamese architecture. We denote by z^j the vector composed of each value at dimension j in all vectors in Z. We define the variance regularization term v as a hinge function on the standard deviation of the embeddings along the batch dimension:

$$v(Z) = \frac{1}{d}\sum_{j=1}^{d} \max(0, \gamma - S(z^j, \epsilon)), \tag{1}$$

where S is the regularized standard deviation defined by:

$$S(x, \epsilon) = \sqrt{\mathrm{Var}(x) + \epsilon}, \tag{2}$$

γ is a constant target value for the standard deviation, fixed to 1 in our experiments, ϵ is a small scalar preventing numerical instabilities. This criterion encourages the variance inside the current

对共享权重和相同结构的要求，为多模态信号的联合嵌入自监督学习打开道路。

信息最大化方法 一种防坍塌原则是最大化嵌入的信息量。近期有 W-MSE（Ermolov et al., 2021）和 Barlow Twins（Zbontar et al., 2021）。W-MSE 的额外模块将嵌入变换到协方差矩阵的特征空间（白化或 Karhunen-Loève 变换），并迫使所得向量在单位球面均匀分布。Barlow Twins 则使两侧嵌入向量的归一化交叉相关矩阵接近单位矩阵。两者都试图让嵌入变量去相关，防止变量携带冗余信息的信息坍塌。所有变量都在批内归一化，因此没有缩小或扩大的动机，这似乎足以防坍塌。我们借鉴 Barlow Twins 的去相关机制，但对两组嵌入各变量加入显式方差保持项，因此无需任何归一化。

4. VICREG：详细描述

VICReg 遵循自监督学习近期趋势（Caron et al., 2020; Grill et al., 2020; Chen & He, 2020; Zbontar et al., 2021; Chen et al., 2020a），采用联合嵌入结构。与许多以往方法不同，结构可完全对称，也可完全非对称，两个分支不共享结构或参数。多数实验采用两分支相同且共享权重的孪生网络。每侧编码器 f_θ 输出用于下游任务的表征，再由扩展器 h_ϕ 映射至计算损失的嵌入空间。扩展器作用有二：（1）消除两种表征的差异信息；（2）以非线性方式扩展维度，使嵌入变量去相关能减少表征变量之间的依赖，而不只是相关性。损失的 s 项学习对数据变换的不变性；方差项 v 防止范数坍塌；协方差项 c 通过去相关各维度，防止信息坍塌。预训练后丢弃扩展器，使用编码器表征完成下游任务。

4.1. 方法

给定从数据集 $\mathcal{D}$ 抽取的图像 i，从分布 $\mathcal{T}$ 抽取变换 t、t'，生成 i 的两个不同视图 $x = t(i)$、$x' = t'(i)$。变换是随机图像裁剪，随后颜色扰动。分布 $\mathcal{T}$ 详见附录C。视图 x、x' 先由 f_θ 编码为表征 $y = f_\theta(x)$、$y' = f_\theta(x')$，再由扩展器 h_ϕ 映射为嵌入 $z = h_\phi(y)$、$z' = h_\phi(y')$。损失在嵌入 z、z' 上计算。

下面描述损失中的方差、不变性和协方差项。图像按批处理，$Z = [z_1, \dots, z_n]$、$Z' = [z'_1, \dots, z'_n]$ 表示孪生结构两侧输出的嵌入批次，每批包含 n 个 d 维向量。z^j 表示将 Z 中所有向量第 j 维的数值组成的向量。方差正则项 v 定义为沿批量维度的嵌入标准差的铰链函数：

$$v(Z) = \frac{1}{d}\sum_{j=1}^{d} \max(0, \gamma - S(z^j, \epsilon)), \tag{1}$$

其中 S 是正则化标准差，定义为：

$$S(x, \epsilon) = \sqrt{\mathrm{Var}(x) + \epsilon}, \tag{2}$$

γ 是标准差的常量目标，实验固定为1；ϵ 是防止数值不稳定的小标量。该准则鼓励当前

batch to be equal to γ along each dimension, preventing collapse with all the inputs mapped on the same vector. Using the standard deviation and not directly the variance is crucial. Indeed, if we take $S(x) = \text{Var}(x)$ in the hinge function, the gradient of S with respect to x becomes close to 0 when x is close to $\bar{x}$. In this case, the gradient of v also becomes close to 0 and the embeddings collapse. We define the covariance matrix of Z as:

$$C(Z) = \frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})(z_i - \bar{z})^T, \quad \text{where} \quad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i. \tag{3}$$

Inspired by Barlow Twins Zbontar et al. (2021), we can then define the covariance regularization term c as the sum of the squared off-diagonal coefficients of $C(Z)$, with a factor $1/d$ that scales the criterion as a function of the dimension:

$$c(Z) = \frac{1}{d}\sum_{i \neq j}[C(Z)]_{i,j}^2. \tag{4}$$

This term encourages the off-diagonal coefficients of $C(Z)$ to be close to 0, decorrelating the different dimensions of the embeddings and preventing them from encoding similar information. Decorrelation at the embedding level ultimately has a decorrelation effect at the representation level, which is a non trivial phenomenon that we study in Appendix D. We finally define the invariance criterion s between Z and Z' as the mean-squared euclidean distance between each pair of vectors, without any normalization:

$$s(Z, Z') = \frac{1}{n}\sum_{i} \|z_i - z'_i\|_2^2. \tag{5}$$

The overall loss function is a weighted average of the invariance, variance and covariance terms:

$$\ell(Z, Z') = \lambda s(Z, Z') + \mu[v(Z) + v(Z')] + \nu[c(Z) + c(Z')], \tag{6}$$

where λ, μ and ν are hyper-parameters controlling the importance of each term in the loss. In our experiments, we set $\nu = 1$ and perform a grid search on the values of λ and μ with the base condition $\lambda = \mu > 1$. The overall objective function taken on all images over an unlabelled dataset $\mathcal{D}$ is given by:

$$\mathcal{L} = \sum_{I \in \mathcal{D}} \sum_{t, t' \sim \mathcal{T}} \ell(Z^I, Z'^I), \tag{7}$$

where Z^I and Z'^I are the batches of embeddings corresponding to the batch of images I transformed by t and t'. The objective is minimized for several epochs, over the encoder parameters θ and expander parameters ϕ. We illustrate the architecture and loss function of VICReg in Figure 1.

4.2 Implementation details

Implementation details for pretraining with VICReg on the 1000-classes ImagetNet dataset without labels are as follows. Coefficients λ and μ are 25 and ν is 1 in Eq. (6), and ϵ is 0.0001 in Eq. (1). We give more details on how we choose the coefficients of the loss function in Appendix D.4. The encoder network f_θ is a standard ResNet-50 backbone He et al. (2016) with 2048 output units. The expander h_ϕ is composed of two fully-connected layers with batch normalization (BN) Ioffe & Szegedy (2015) and ReLU, and a third linear layer. The sizes of all 3 layers were set to 8192. As with Barlow Twins, performance improves when the size of the expander layers is larger than the dimension of the representation. The impact of the expander dimension on performance is studied in Appendix D. The training protocol follows those of BYOL and Barlow Twins: LARS optimizer You et al. (2017); Goyal et al. (2017) run for 1000 epochs with a weight decay of 10^{-6} and a learning rate $lr = batch_size/256 \times base_lr$, where $batch_size$ is set to 2048 by default and $base_lr$ is a base learning rate set to 0.2. The learning rate follows a cosine decay schedule Loshchilov & Hutter (2017), starting from 0 with 10 warmup epochs and with final value of 0.002.

5 Results

In this section, we evaluate the representations obtained after self-supervised pretraining of a ResNet-50 He et al. (2016) backbone with VICReg during 1000 epochs, on the training set of ImageNet, using the training protocol described in section 4. We also pretrain on pairs of image and text data and evaluate on retrieval tasks on the MS-COCO dataset.

批次每个维度的值达到 γ，防止全部输入映射到同一向量的坍塌。使用标准差而非直接使用方差至关重要。如果铰链函数中采用 $S(x) = \text{Var}(x)$，则当 x 接近 $\bar{x}$ 时，S 关于 x 的梯度趋近0，v 的梯度也趋近0，导致嵌入坍塌。将 Z 的协方差矩阵定义为：

$$C(Z) = \frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})(z_i - \bar{z})^T, \quad \text{where} \quad \bar{z} = \frac{1}{n}\sum_{i=1}^{n} z_i. \tag{3}$$

受 Barlow Twins（Zbontar et al., 2021）启发，协方差正则项 c 定义为 $C(Z)$ 非对角系数的平方和，乘以 $1/d$ 以按维度缩放：

$$c(Z) = \frac{1}{d}\sum_{i \neq j}[C(Z)]_{i,j}^2. \tag{4}$$

该项使 $C(Z)$ 的非对角系数接近0，令嵌入不同维度去相关，防止编码相似信息。嵌入层面的去相关最终也影响表征层面，这是非平凡现象，详见附录D。最后，将 Z、Z' 的不变性准则 s 定义为每对向量间的平方欧氏距离均值，不进行任何归一化：

$$s(Z, Z') = \frac{1}{n}\sum_{i} \|z_i - z'_i\|_2^2. \tag{5}$$

总损失是不变性、方差和协方差项的加权平均：

$$\ell(Z, Z') = \lambda s(Z, Z') + \mu[v(Z) + v(Z')] + \nu[c(Z) + c(Z')], \tag{6}$$

其中 λ、μ、ν 是控制各项重要性的超参数。实验设定 $\nu = 1$，并在基本条件 $\lambda = \mu > 1$ 下对 λ、μ 网格搜索。无标注数据集 $\mathcal{D}$ 全部图像上的总目标为：

$$\mathcal{L} = \sum_{I \in \mathcal{D}} \sum_{t, t' \sim \mathcal{T}} \ell(Z^I, Z'^I), \tag{7}$$

其中 Z^I、Z'^I 是一批图像 I 经 t、t' 变换所对应的嵌入批次。对编码器参数 θ、扩展器参数 ϕ 优化该目标若干轮。结构及损失见图1。

4.2. 实现细节

在无标注1000类 ImageNet 上使用 VICReg 预训练的细节如下：式6的系数 λ、μ 为25，ν 为1；式1的 ϵ 为0.0001。损失系数选择详见附录D.4。编码器 f_θ 是标准 ResNet-50（He et al., 2016），输出2048维。扩展器 h_ϕ 包含两个带批归一化（BN；Ioffe & Szegedy, 2015）和 ReLU 的全连接层，以及第三个线性层；三层均为8192维。与 Barlow Twins 一样，扩展器层宽大于表征维度时性能更好；维度影响见附录D。训练遵循 BYOL 和 Barlow Twins：LARS 优化器（You et al., 2017; Goyal et al., 2017），1000轮，权重衰减 10^{-6}，学习率 $lr = batch_size/256 \times base_lr$；$batch_size$ 默认2048，基础学习率 $base_lr$ 为0.2。采用余弦衰减（Loshchilov & Hutter, 2017），从0开始预热10轮，最终降至0.002。

5. 结果

本节评估按照第4节流程，在 ImageNet 训练集上用 VICReg 自监督预训练1000轮的 ResNet-50（He et al., 2016）表征。还在图像—文本对上预训练，在 MS-COCO 检索任务上评估。

Table 1: **Evaluation on ImageNet.** Evaluation of the representations obtained with a ResNet-50 backbone pretrained with VICReg on: (1) linear classification on top of the frozen representations from ImageNet; (2) semi-supervised classification on top of the fine-tuned representations from 1% and 10% of ImageNet samples. We report Top-1 and Top-5 accuracies (in %). Top-3 best self-supervised methods are underlined.

Method	Linear		Semi-supervised			
	Top-1	Top-5	Top-1		Top-5	
			1%	10%	1%	10%
Supervised	76.5	-	25.4	56.4	48.4	80.4
MoCo He et al. (2020)	60.6	-	-	-	-	-
PIRL Misra & Maaten (2020)	63.6	-	-	-	57.2	83.8
CPC v2 Hénaff et al. (2019)	63.8	-	-	-	-	-
CMC Tian et al. (2019)	66.2	-	-	-	-	-
SimCLR Chen et al. (2020a)	69.3	89.0	48.3	65.6	75.5	87.8
MoCo v2 Chen et al. (2020c)	71.1	-	-	-	-	-
SimSiam Chen & He (2020)	71.3	-	-	-	-	-
SwAV Caron et al. (2020)	71.8	-	-	-	-	-
InfoMin Aug Tian et al. (2020)	73.0	91.1	-	-	-	-
OBoW Gidaris et al. (2021)	73.8	-	-	-	82.9	90.7
BYOL Grill et al. (2020)	74.3	91.6	53.2	68.8	78.4	89.0
SwAV (w/ multi-crop) Caron et al. (2020)	75.3	-	53.9	70.2	78.5	89.9
Barlow Twins Zbontar et al. (2021)	73.2	91.0	55.0	69.7	79.2	89.3
VICReg (ours)	73.2	91.1	54.8	69.5	79.4	89.5

5.1 Evaluation on ImageNet

Following the ImageNet Deng et al. (2009) linear evaluation protocol, we train a linear classifier on top of the frozen representations of the ResNet-50 backbone pretrained with VICReg. We also evaluate the performance of the backbone when fine-tuned with a linear classifier on a subset of ImageNet's training set using 1% or 10% of the labels, using the split of Chen et al. (2020a). We give implementation details about the optimization procedure for these tasks in Appendix C. We have applied the training procedure described in section 4 with three different random initialization. The numbers reported in Table 1 for VICReg are the mean scores, and we have observed that the difference between worse and best run is lower than 0.1% accuracy for linear classification, which shows that VICReg is a very stable algorithm. Lack of time has prevented us from doing the same for the semi-supervised classification experiments, and the experiments of section 5.2 and 6, but we expect similar conclusion to hold. We compare in Table 1 our results on both tasks against other methods on the validation set of ImageNet. The performance of VICReg is on par with the state of the art without using the negative pairs of SimCLR, the clusters of SwAV, the bag-of-words representations of OBoW, or any asymmetric networks architectural tricks such as the momentum encoder of BYOL and the stop-gradient operation of SimSiam. The performance is comparable to that of Barlow Twins, which shows that VICReg's more explicit way of constraining the variance and comparing views has the same power than maximizing cross-correlations between pairs of twin dimensions. The main advantage of VICReg is the modularity of its objective function and the applicability to multi-modal setups.

5.2 Transfer to other downstream tasks

Following the setup from Misra & Maaten (2020), we train a linear classifier on top of the frozen representations learnt by our pretrained ResNet-50 backbone on a variety of different datasets: the Places205 Zhou et al. (2014) scene classification dataset, the VOC07 Everingham et al. (2010) multi-label image classification dataset and the iNaturalist2018 Horn et al. (2018) fine-grained image classification dataset. We then evaluate the quality of the representations by transferring to other vision tasks including VOC07+12 Everingham et al. (2010) object detection using Faster R-CNN Ren et al. (2015) with a R50-C4 backbone, and COCO Lin et al. (2014) instance segmentation using Mask-R-CNN He et al. (2017) with a R50-FPN backbone. We report the performance in Table 2,

表1：ImageNet 评估。使用 VICReg 预训练 ResNet-50 的表征：（1）ImageNet 冻结表征上的线性分类；（2）用1%及10% ImageNet 样本微调表征的半监督分类。报告 top-1、top-5 准确率（%），前三名自监督方法加下划线。

方法	线性		半监督			
	Top-1	Top-5	Top-1		Top-5	
			1%	10%	1%	10%
监督学习	76.5	-	25.4	56.4	48.4	80.4
MoCo He et al. (2020)	60.6	-	-	-	-	-
PIRL Misra & Maaten (2020)	63.6	-	-	-	57.2	83.8
CPC v2 Hénaff et al. (2019)	63.8	-	-	-	-	-
CMC Tian et al. (2019)	66.2	-	-	-	-	-
SimCLR Chen et al. (2020a)	69.3	89.0	48.3	65.6	75.5	87.8
MoCo v2 Chen et al. (2020c)	71.1	-	-	-	-	-
SimSiam Chen & He (2020)	71.3	-	-	-	-	-
SwAV Caron et al. (2020)	71.8	-	-	-	-	-
InfoMin Aug Tian et al. (2020)	73.0	91.1	-	-	-	-
OBoW Gidaris et al. (2021)	73.8	-	-	-	82.9	90.7
BYOL Grill et al. (2020)	74.3	91.6	53.2	68.8	78.4	89.0
SwAV（多裁剪）Caron et al.（2020）	75.3	-	53.9	70.2	78.5	89.9
Barlow Twins Zbontar et al. (2021)	73.2	91.0	55.0	69.7	79.2	89.3
VICReg（本文）	73.2	91.1	54.8	69.5	79.4	89.5

5.1. ImageNet 评估

遵循 ImageNet（Deng et al., 2009）线性评估流程，在 VICReg 预训练 ResNet-50 主干的冻结表征上训练线性分类器。还采用 Chen et al.（2020a）的划分，使用1%或10%标注，在 ImageNet 训练子集上联合微调主干和线性分类器。优化细节见附录C。第4节的训练流程使用三种不同随机初始化。表1的 VICReg 数字为平均分；在线性分类中，最好与最差运行的准确率差小于0.1个百分点，说明算法非常稳定。时间限制使我们未能对半监督分类及第5.2、6节实验重复同样评估，但预期结论类似。表1将两个任务的 ImageNet 验证集结果与其他方法比较。VICReg 与最优方法相当，无需 SimCLR 的负样本对、SwAV 的聚类、OBoW 的词袋表征，或 BYOL 的动量编码器、SimSiam 的停止梯度等非对称结构技巧。它与 Barlow Twins 性能接近，表明显式约束方差和比较视图，与最大化双网络对应维度交叉相关同样有效。VICReg 的主要优势是目标函数的模块化，以及对多模态设置的适用性。

5.2. 向其他下游任务迁移

遵循 Misra & Maaten（2020），在预训练 ResNet-50 主干的冻结表征上，针对不同数据集训练线性分类器：Places205（Zhou et al., 2014）场景分类、VOC07（Everingham et al., 2010）多标签分类，以及 iNaturalist2018（Horn et al., 2018）细粒度分类。再通过迁移到其他视觉任务评估表征：VOC07+12（Everingham et al., 2010）用 R50-C4 主干的 Faster R-CNN（Ren et al., 2015）目标检测；COCO（Lin et al., 2014）用 R50-FPN 主干的 Mask R-CNN（He et al., 2017）实例分割。表2报告性能，

Table 2: **Transfer learning on downstream tasks.** Evaluation of the representations from a ResNet-50 backbone pretrained with VICReg on: (1) linear classification tasks on top of frozen representations, we report Top-1 accuracy (in %) for Places205 Zhou et al. (2014) and iNat18 Horn et al. (2018), and mAP for VOC07 Everingham et al. (2010); (2) object detection with fine-tunning, we report AP_{50} for VOC07+12 using Faster R-CNN with C4 backbone Ren et al. (2015); (3) object detection and instance segmentation, we report AP for COCO Lin et al. (2014) using Mask R-CNN with FPN backbone He et al. (2017). We use † to denote the experiments run by us. Top-3 best self-supervised methods are underlined.

	Linear Classification			Object Detection		
Method	Places205	VOC07	iNat18	VOC07+12	COCO det	COCO seg
Supervised	53.2	87.5	46.7	81.3	39.0	35.4
MoCo He et al. (2020)	46.9	79.8	31.5	-	-	-
PIRL Misra & Maaten (2020)	49.8	81.1	34.1	-	-	-
SimCLR Chen et al. (2020a)	52.5	85.5	37.2	-	-	-
MoCo v2 Chen et al. (2020c)	51.8	86.4	38.6	82.5	39.8	36.1
SimSiam Chen & He (2020)	-	-	-	82.4	-	-
BYOL Grill et al. (2020)	54.0	86.6	47.6	-	40.4†	37.0†
SwAV (m-c) Caron et al. (2020)	56.7	88.9	48.6	82.6	41.6	37.8
OBoW Gidaris et al. (2021)	56.8	89.3	-	82.9	-	-
Barlow Twins Grill et al. (2020)	54.1	86.2	46.5	82.6	40.0†	36.7†
VICReg (ours)	54.3	86.6	47.0	82.4	39.4	36.4

Table 3: **Evaluation on MS-COCO 5K retrieval tasks.** Comparison of VICReg with the contrastive loss of VSE++ Faghri et al. (2018), and with Barlow Twins, pretrain on the training set of MS-COCO. In all settings, the encoder for text is a word embedding followed by a GRU layer, the encoder for images is a ResNet-152.

Method	Image-to-text			Text-to-Image		
	R@1	R@5	R@10	R@1	R@5	R@10
Contrastive (VSE++)	30.3	59.4	72.4	41.3	71.1	81.2
Barlow Twins	31.4	60,4	75.1	42.9	74.0	83.5
VICReg	33.6	62.7	77.9	45.2	76.1	84.2

VICReg performs on par with most concurrent methods, and better than Barlow Twins, across all classification tasks, but is slightly behind the top-3 on detection tasks.

5.3 Multi-modal pretraining on MS-COCO

One fundamental difference of VICReg compared to Barlow Twins is the way the branches are regularized. In VICReg, both branches are regularized independently, as the covariance term is applied on each branch separately, which works better in the scenarios where the branches are completely different, have different types of architecture and process different types of data. Indeed, the statistics of the output of the two branches can be very different, and the amount of regularization required for each may vary a lot. In Barlow Twins, the regularization is applied on the cross-correlation matrix, which favors the scenarios where the branches produce outputs with similar statistics. We demonstrate the capabilities of VICReg in a multi-modal experiment where we pretrain on pairs of images and corresponding captions on the MS-COCO dataset. We regularize each branch with a different coefficient, which is not possible with Barlow Twins, and we show that VICReg outperforms Barlow Twins on image and text retrieval downstream tasks. Table 3 reports the performance of VICReg against the contrastive loss proposed by VSE++ Faghri et al. (2018), and against Barlow Twins, in the identical setting proposed in Faghri et al. (2018). VICReg outperforms the two by a significant margin.

表2：下游任务迁移学习。评估 VICReg 预训练 ResNet-50 的表征：（1）冻结表征上的线性分类，报告 Places205（Zhou et al., 2014）、iNat18（Horn et al., 2018）的 top-1 准确率（%），VOC07（Everingham et al., 2010）的 mAP；（2）微调目标检测，用 C4 主干 Faster R-CNN（Ren et al., 2015）报告 VOC07+12 的 AP_{50}；（3）目标检测与实例分割，用 FPN 主干 Mask R-CNN（He et al., 2017）报告 COCO（Lin et al., 2014）的 AP。† 表示我们运行的实验，前三名自监督方法加下划线。

	线性分类			目标检测		
方法	Places205	VOC07	iNat18	VOC07+12	COCO 检测	COCO 分割
监督学习	53.2	87.5	46.7	81.3	39.0	35.4
MoCo He et al. (2020)	46.9	79.8	31.5	-	-	-
PIRL Misra & Maaten (2020)	49.8	81.1	34.1	-	-	-
SimCLR Chen et al. (2020a)	52.5	85.5	37.2	-	-	-
MoCo v2 Chen et al. (2020c)	51.8	86.4	38.6	82.5	39.8	36.1
SimSiam Chen & He (2020)	-	-	-	82.4	-	-
BYOL Grill et al. (2020)	54.0	86.6	47.6	-	40.4†	37.0†
SwAV（多裁剪）Caron et al.（2020）	56.7	88.9	48.6	82.6	41.6	37.8
OBoW Gidaris et al. (2021)	56.8	89.3	-	82.9	-	-
Barlow Twins Grill et al. (2020)	54.1	86.2	46.5	82.6	40.0†	36.7†
VICReg（本文）	54.3	86.6	47.0	82.4	39.4	36.4

表3：MS-COCO 5K 检索评估。在 MS-COCO 训练集预训练，比较 VICReg、VSE++（Faghri et al., 2018）对比损失与 Barlow Twins。全部设置中，文本编码器为词嵌入加 GRU 层，图像编码器为 ResNet-152。

方法	图像到文本			文本到图像		
	R@1	R@5	R@10	R@1	R@5	R@10
对比学习（VSE++）	30.3	59.4	72.4	41.3	71.1	81.2
Barlow Twins	31.4	60,4	75.1	42.9	74.0	83.5
VICReg	33.6	62.7	77.9	45.2	76.1	84.2

在所有分类任务上，VICReg 与多数同期方法相当，并优于 Barlow Twins；但在检测任务上略低于前三名。

5.3. MS-COCO 多模态预训练

VICReg 与 Barlow Twins 的根本差异之一是分支正则化方式。VICReg 将协方差项分别用于各分支，独立正则化，因此当分支完全不同、采用不同结构或处理不同数据时更好。两侧输出的统计量可能差别很大，所需正则化强度也可能不同。Barlow Twins 在交叉相关矩阵上正则化，更适合两侧输出统计量相似的情形。我们在 MS-COCO 图像—对应描述对上进行多模态预训练，展示 VICReg 的能力：对各分支使用不同系数，这在 Barlow Twins 中无法做到。VICReg 在下游图像与文本检索中优于 Barlow Twins。表3采用 Faghri et al.（2018）的相同设置，比较 VICReg、VSE++ 的对比损失及 Barlow Twins；VICReg 明显优于两者。

Table 4: **Effect of incorporating variance and covariance regularization in different methods.** Top-1 ImageNet accuracy with the linear evaluation protocol after 100 pretraining epochs. For all methods, pretraining follows the architecture, the optimization and the data augmentation protocol of the original method using our reimplementation. ME: Momentum Encoder. SG: stop-gradient. PR: predictor. BN: Batch normalization layers after input and inner linear layers in the expander. No Reg: No additional regularization. Var Reg: Variance regularization. Var/Cov Reg: Variance and Covariance regularization. Unmodified original setups are marked by a †.

Method	ME	SG	PR	BN	No Reg	Var Reg	Var/Cov Reg
BYOL	✓	✓	✓	✓	69.3†	70.2	69.5
SimSiam		✓	✓	✓	67.9†	68.1	67.6
SimSiam		✓	✓		35.1	67.3	67.1
SimSiam		✓			collapse	56.8	66.1
VICReg			✓		collapse	56.2	67.3
VICReg			✓	✓	collapse	57.1	68.7
VICReg				✓	collapse	57.5	68.6†
VICReg					collapse	56.5	67.4

6 ANALYSIS

In this section we study how the different components of our method contribute to its performance, as well as how they interact with components from other self-supervised methods. We also evaluate different scenarios where the branches have different weights and architecture. All reported results are obtained on the linear evaluation protocol, using a ResNet-50 backbone if not mentioned otherwise, and 100 epochs of pretraining, which gives results consistent with those obtained with 1000 epochs of pretraining. The optimization setting used for each experiment is described in Appendix C.

Asymmetric networks. We study the impact of different components used in asymmetric architectures and the effects of adding variance and covariance regularization, in terms of performance and training stability. Starting from a simple symmetric architecture with an encoder and an expander without batch normalization, which correspond to VICReg without batch normalization in the expander, we progressively add batch normalization in the inner layers of the expander, a predictor, a stop-gradient operation and a momentum encoder. We use the training protocol and architecture of SimSiam Chen & He (2020) when a stop-gradient is used and the training protocol and architecture of BYOL Grill et al. (2020) when a momentum encoder is used. The predictor as used in SimSiam and BYOL is a learnable module g_ψ that predicts the embedding of a view given the embedding of the other view of the same image. If z and z' are the embeddings of two views of an image, then $p = g_\psi(z)$ and $p' = g_\psi(z')$ are the predictions of each view. The invariance loss function of Eq. (5) is now computed between a batch of embeddings $Z = [z_1, \dots, z_n]$ and the corresponding batch of predictions $P = [p'_1, \dots, p'_n]$, then symmetrized:

$$s(Z, Z', P, P') = \frac{1}{2n} \sum_i D(z_i - p'_i) + \frac{1}{2n} \sum_i D(z'_i - p_i), \tag{8}$$

where D is a distance function that depends on the method used. BYOL uses the mean square error between l_2-normalized vectors, SimSiam uses the negative cosine similarity loss and VICReg uses the mean square error without l_2-normalization. The variance and covariance terms are regularizing the output Z and Z' of the expander, which we empirically found to work better than regularizing the output of the predictor. We compare different settings in Table 4, based on the default data augmentation, optimization and architecture settings of the original BYOL, SimSiam and VICReg methods. In all settings, the absence of BN indicates that BN is also removed in the predictor when one is used.

We analyse first the impact of variance regularization (VR) in the different settings. When using VR, adding a predictor (PR) to VICReg does not lead to a significant change of the performance, which indicates that PR is redundant with VR. In comparison, without VR, the representations collapse, and both stop-gradient (SG) and PR are necessary. Batch normalization in the inner layers of the expander (BN) in VICReg leads to a 1.0% increase in the performance, which is not a big improvement considering that SG and PR without BN is performing very poorly at 35.1%.

表4：不同方法加入方差和协方差正则化的影响。预训练100轮后的 ImageNet 线性 top-1 准确率。全部方法使用我们的复现，遵循原方法结构、优化和增强流程。ME：动量编码器；SG：停止梯度；PR：预测器；BN：扩展器输入及内部线性层后的批归一化；No Reg：无额外正则；Var Reg：方差正则；Var/Cov Reg：方差与协方差正则。未修改的原始设置用 † 标记。

方法	ME	SG	PR	BN	无正则	方差正则	方差／协方差正则
BYOL	✓	✓	✓	✓	69.3†	70.2	69.5
SimSiam		✓	✓	✓	67.9†	68.1	67.6
SimSiam		✓	✓		35.1	67.3	67.1
SimSiam		✓			坍塌	56.8	66.1
VICReg			✓		坍塌	56.2	67.3
VICReg			✓	✓	坍塌	57.1	68.7
VICReg				✓	坍塌	57.5	68.6†
VICReg					坍塌	56.5	67.4

6. 分析

本节研究各组件如何贡献性能，以及它们与其他自监督组件如何交互；还评估分支不同权重与结构的情形。全部结果使用线性评估，未另说明时采用 ResNet-50，预训练100轮；结果与1000轮的结论一致。各实验优化设置见附录C。

非对称网络 我们研究非对称结构中不同组件，以及加入方差和协方差正则化对性能和训练稳定性的影响。从包含编码器和无 BN 扩展器的简单对称结构开始，即扩展器不使用 BN 的 VICReg，逐步加入扩展器内部 BN、预测器、停止梯度及动量编码器。使用停止梯度时采用 SimSiam（Chen & He, 2020）的训练流程与结构；使用动量编码器时采用 BYOL（Grill et al., 2020）。预测器是可训练模块 g_ψ，给定同一图像一个视图的嵌入，预测另一视图的嵌入。若 z、z' 是两个视图嵌入，$p = g_\psi(z)$ 和 $p' = g_\psi(z')$ 为各视图的预测。式5不变性损失改为在一批嵌入 $Z = [z_1, \dots, z_n]$ 与对应预测 $P = [p'_1, \dots, p'_n]$ 之间计算，然后对称化：

$$s(Z, Z', P, P') = \frac{1}{2n} \sum_i D(z_i - p'_i) + \frac{1}{2n} \sum_i D(z'_i - p_i), \tag{8}$$

其中 D 是依方法而定的距离函数。BYOL 使用 l_2 归一化向量的均方误差，SimSiam 使用负余弦相似度，VICReg 使用不含 l_2 归一化的均方误差。方差与协方差项正则化扩展器输出 Z、Z'；实验发现优于正则化预测器输出。表4基于原 BYOL、SimSiam、VICReg 的默认增强、优化及结构，比较不同设置。所有设置中，“无 BN”也表示预测器若存在则去除其 BN。

首先分析不同设置中的方差正则化（VR）。使用 VR 时，给 VICReg 加预测器（PR）并不显著改变性能，说明 PR 与 VR 冗余。不用 VR 时，表征会坍塌，停止梯度（SG）和 PR 都必要。VICReg 扩展器内部 BN 提高性能1.0个百分点；这一改善并不大，考虑到没有 BN 的 SG 加 PR 仅达到 35.1%，表现很差。

Table 5: **Impact of sharing weights or not between branches.** Top-1 accuracy on linear classification with 100 pretraining epochs. The encoder and expander of both branches can share the same architecture and share their weights (SW), share the same architecture with different weights (DW), or have different architectures (DA). The encoders can be ResNet-50, ResNet-101 or ViT-S.

	SW R50	DW R50	DA R50/R101	DA R50/ViT-S
BYOL	69.3	✗	✗	✗
SimCLR	64.4	63.1	63.9	63.5
Barlow Twins	68.7	64.2	65.3	63.9
VICReg	68.6	66.5	68.1	66.2

Finally, incorporating VR with SG or ME further improves the performance by small margins of respectively 0.2% and 0.9%, which might be explained by the fact that these architectural tricks that prevent collapse are not perfectly maintaining the variance of the representations, i.e. very slow collapse is happening with these methods. We explain this intuition by studying the evolution of the standard deviation of the representations during pretraining for BYOL and SimSiam in Appendix D. We then analyse the impact of adding additional covariance regularization (CR) in the different settings, along with variance regularization. We found that optimization with SG and CR is hard, even if our analysis of the average correlation coefficient of the representations during pretraining in Appendix D shows that both fulfill the same objective.

The performance of BYOL and SimSiam slightly drops compared to VR only, except when PR is removed, where SG becomes useless. BN is still useful and improves the performance by 1.3%. Finally with CR, PR does not harm the performance and even improves it by a very small margin. VICReg+PR with 1000 epochs of pretraining exactly matches the score of VICReg (73.2% on linear classification).

Weight sharing. Contrary to most self-supervised learning approaches based on Siamese architectures, VICReg has several unique properties: (1) weights do not need to be shared between the branches, each branch's weights are updated independently of the other branch's weights; (2) the branches are regularized independently, the variance and covariance terms are computed on each branch individually; (3) no predictor is necessary unlike with methods where one branch predicts outputs of the other branch. We compare the robustness of VICReg against other methods in different scenarios where the weights of the branches can be shared (SW), not shared (DW), and where the encoders can have different architectures (DA). Among other self-supervised methods, SimCLR and Barlow Twins are the only ones that can handle these scenarios. The asymmetric methods that are based on a discrepancy between the branches requires either the architecture or the weights to be shared between the branches. The performance drops by 2.1% with VICReg and 4.5% with Barlow Twins, between the shared weights scenario (SW) and the different weight scenario (DW). The difference between VICReg and Barlow Twins is also significant in scenarios with different architectures, in particular VICReg performs better than Barlow Twins by 2.8% with ResNet-50/ResNet-101 and better by 2.3% with ResNet-50/ViT-S Dosovitskiy et al. (2021). This shows that VICReg is more robust than Barlow Twins in these kind of scenarios. The performance of SimCLR remains stable across scenarios, but is significantly worse than the performance of VICReg. Importantly, **the ability of VICReg to function with different parameters, architectures, and input modalities for the branches widens the applicability to joint-embedding SSL to many applications, including multi-modal signals.**

7 Conclusion

We introduced VICReg, a simple approach to self-supervised learning based on a triple objective: learning invariance to different views with a invariance term, avoiding collapse of the representations with a variance preservation term, and maximizing the information content of the representation with a covariance regularization term. VICReg achieves results on par with the state of the art on many downstream tasks, but is not subject to the same limitations as most other methods, particularly because it does not require the embedding branches to be identical or even similar.

表5：分支共享权重的影响。预训练100轮的线性分类 top-1 准确率。两分支编码器和扩展器可采用相同结构及共享权重（SW）、相同结构不同权重（DW），或不同结构（DA）。编码器可为 ResNet-50、ResNet-101 或 ViT-S。

	SW R50	DW R50	DA R50/R101	DA R50/ViT-S
BYOL	69.3	✗	✗	✗
SimCLR	64.4	63.1	63.9	63.5
Barlow Twins	68.7	64.2	65.3	63.9
VICReg	68.6	66.5	68.1	66.2

最后，在 SG 或 ME 中加入 VR，分别再提升0.2、0.9个百分点。可能原因是防坍塌结构技巧并未完全保持表征方差，即发生了很慢的坍塌。附录D通过分析 BYOL、SimSiam 预训练表征标准差的变化解释这一直觉。随后在不同设置中，分析方差正则化之外再加入协方差正则化（CR）的影响。发现 SG 与 CR 联合优化较困难，尽管附录D的预训练平均相关系数分析显示两者目的相同。

BYOL 和 SimSiam 相比仅加 VR 时性能略降；移除 PR 的情况例外，此时 SG 无用。BN 仍有效，提升1.3个百分点。最后，加入 CR 时，PR 不损害性能，甚至略有改善。VICReg+PR 预训练1000轮，与 VICReg 得分完全相同（线性分类73.2%）。

权重共享 与多数基于孪生结构的自监督方法不同，VICReg 有以下独特性质：（1）分支无需共享权重，各自独立更新；（2）分支独立正则化，方差与协方差项分别计算；（3）无需预测器，而某些方法需要一个分支预测另一分支的输出。我们在分支共享权重（SW）、不同权重（DW）以及编码器不同结构（DA）的情况下，比较 VICReg 与其他方法的鲁棒性。其他自监督方法中，只有 SimCLR 与 Barlow Twins 能处理这些情形。基于分支差异的非对称方法，要求分支共享结构或权重。从 SW 到 DW，VICReg 性能下降2.1个百分点，Barlow Twins 下降4.5个百分点。不同结构时，差异也很明显：ResNet-50／ResNet-101 组合中，VICReg 比 Barlow Twins 高2.8个百分点；ResNet-50／ViT-S（Dosovitskiy et al., 2021）组合中高2.3个百分点。这说明 VICReg 在这些情况下更鲁棒。SimCLR 在各情形中性能稳定，但明显低于 VICReg。VICReg 允许分支采用不同参数、结构和输入模态，使联合嵌入自监督学习适用于更多应用，包括多模态信号。

7. 结论

我们提出 VICReg，一种简单自监督方法，包含三个目标：不变性项学习不同视图的不变性，方差保持项防止表征坍塌，协方差正则项最大化表征信息量。它在许多下游任务上与最优方法相当，却不受多数方法同样的限制，尤其无需嵌入分支相同，甚至无需相似。

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REFERENCES

Yuki Markus Asano, Christian Rupprecht, and Andrea Vedaldi. Self-labelling via simultaneous clustering and representation learning. In *ICLR*, 2020. 3

Philip Bachman, R Devon Hjelm, and William Buchwalter. Learning representations by maximizing mutual information across views. In *NeurIPS*, 2019. 1

Miguel A. Bautista, Artsiom Sanakoyeu, Ekaterina Sutter, and Björn Ommer. Cliquecnn: Deep unsupervised exemplar learning. In *NeurIPS*, 2016. 3

Jane Bromley, Isabelle Guyon, Yann LeCun, Eduard Sackinger, and Roopak Shah. Signature verification using a "siamese" time delay neural network. In *NeurIPS*, 1994. 1, 3

Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze. Deep clustering for unsupervised learning. In *ECCV*, 2018. 1, 3

Mathilde Caron, Piotr Bojanowski, Julien Mairal, and Armand Joulin. Unsupervised pre-training of image features on non-curated data. In *ICCV*, 2019. 3

Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. In *NeurIPS*, 2020. 1, 3, 4, 6, 7, 14, 16, 17, 18, 20

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey E. Hinton. A simple framework for contrastive learning of visual representations. In *ICML*, 2020a. 1, 3, 4, 6, 7, 14, 16, 17, 20, 21

Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big self-supervised models are strong semi-supervised learners. In *NeurIPS*, 2020b. 3

Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In *CVPR*, 2020. 1, 3, 4, 6, 7, 8, 14, 16, 17, 21

Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. *arXiv preprint arXiv:2003.04297*, 2020c. 3, 6, 7

Sumit Chopra, Raia Hadsell, and Yann LeCun. Learning a similarity metric discriminatively, with application to face verification. In *CVPR*, 2005. 1

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *NeurIPS*, 2013. 3

Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009. 3, 6

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 9

Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning, 2021. 2, 3, 4, 14

Mark Everingham, Luc Van Gool, John Winn Christopher K. I. Williams, and Andrew Zisserman. The pascal visual object classes (voc) challenge. *IJCV*, 2010. 6, 7, 16

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参考文献

Yuki Markus Asano, Christian Rupprecht, and Andrea Vedaldi. Self-labelling via simultaneous clustering and representation learning. In ICLR, 2020. 3 Philip Bachman, R Devon Hjelm, and William Buchwalter. Learning representations by maximizing mutual information across views. In NeurIPS, 2019. 1 Miguel A. Bautista, Artsiom Sanakoyeu, Ekaterina Sutter, and Björn Ommer. Cliquecnn: Deep unsupervised exemplar learning. In NeurIPS, 2016. 3 Jane Bromley, Isabelle Guyon, Yann LeCun, Eduard Sackinger, and Roopak Shah. Signature verification using a " siamese" time delay neural network. In NeurIPS, 1994. 1, 3 Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Matthijs Douze. Deep clustering for unsuper- vised learning. In ECCV, 2018. 1, 3 Mathilde Caron, Piotr Bojanowski, Julien Mairal, and Armand Joulin. Unsupervised pre-training of image features on non-curated data. In ICCV, 2019. 3 Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. In NeurIPS, 2020. 1, 3, 4, 6, 7, 14, 16, 17, 18, 20 Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey E. Hinton. A simple framework for contrastive learning of visual representations. In ICML, 2020a. 1, 3, 4, 6, 7, 14, 16, 17, 20, 21 Ting Chen, Simon Kornblith, Kevin Swersky, Mohammad Norouzi, and Geoffrey Hinton. Big self-supervised models are strong semi-supervised learners. In NeurIPS, 2020b. 3 Xinlei Chen and Kaiming He. Exploring simple siamese representation learning. In CVPR, 2020. 1, 3, 4, 6, 7, 8, 14, 16, 17, 21 Xinlei Chen, Haoqi Fan, Ross Girshick, and Kaiming He. Improved baselines with momentum contrastive learning. arXiv preprint arXiv:2003.04297, 2020c. 3, 6, 7 Sumit Chopra, Raia Hadsell, and Yann LeCun. Learning a similarity metric discriminatively, with application to face verification. In CVPR, 2005. 1 Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In NeurIPS, 2013. 3 Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In CVPR, 2009. 3, 6 Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In ICLR, 2021. 9 Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self- supervised representation learning, 2021. 2, 3, 4, 14 Mark Everingham, Luc Van Gool, John Winn Christopher K. I. Williams, and Andrew Zisserman. The pascal visual object classes (voc) challenge. IJCV, 2010. 6, 7, 16

Fartash Faghri, David J. Fleet, Jamie Ryan Kiros, and Sanja Fidler. Vse++: Improving visual-semantic embeddings with hard negatives. In *BMVC*, 2018. 7

Rong-En Fan, Kai-Wei Chang, Cho-Jui Hsieh, Xiang-Rui Wang, and Chih-Jen Lin. Liblinear: A library for large linear classification. *JMLR*, 2008. 16

Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick Pérez, and Matthieu Cord. Learning representations by predicting bags of visual words. In *CVPR*, 2020. 3

Spyros Gidaris, Andrei Bursuc, Gilles Puy, Nikos Komodakis, Matthieu Cord, and Patrick Pérez. Online bag-of-visual-words generation for unsupervised representation learning. In *CVPR*, 2021. 1, 3, 6, 7, 14

Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch sgd: Training imagenet in 1 hour. *arXiv preprint arXiv:1706.02677*, 2017. 5

Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Ishan Misra. Vissl. `https://github.com/facebookresearch/vissl`, 2021. 16

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *NeurIPS*, 2020. 1, 3, 4, 6, 7, 8, 14, 16, 17, 20, 21

Raia Hadsell, Sumit Chopra, and Yann LeCun. Dimensionality reduction by learning an invariant mapping. In *CVPR*, 2006. 3

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016. 5

Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In *ICCV*, 2017. 6, 7

Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *CVPR*, 2020. 1, 3, 6, 7, 16

Geoffrey Hinton, Oriol Vinyals, and Jeffrey Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning and Representation Learning Workshop*, 2015. 3

R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Adam Trischler, and Yoshua Bengio. Learning deep representations by mutual information estimation and maximization. In *ICLR*, 2019. 1, 3

Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and detection dataset. In *CVPR*, 2018. 6, 7, 16

Jiabo Huang, Qi Dong andShaogang Gong, and Xiatian Zhu. Unsupervised deep learning by neighbourhood discovery. In *ICML*, 2019. 3

Olivier J. Hénaff, Aravind Srinivas, Jeffrey De Fauw, Ali Razavi, Carl Doersch, S. M. Ali Eslami, and Aäron van den Oord. Data-efficient image recognition with contrastive predictive coding. In *ICML*, 2019. 6

Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *ICML*, 2015. 5

Junnan Li, Pan Zhou, Caiming Xiong, and Steven C.H. Hoi. Prototypical contrastive learning of unsupervised representations. In *ICLR*, 2021. 20

Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context. In *ECCV*, 2014. 6, 7

Fartash Faghri, David J. Fleet, Jamie Ryan Kiros, and Sanja Fidler. Vse++: Improving visual-semantic embeddings with hard negatives. In BMVC, 2018. 7 Rong-En Fan, Kai-Wei Chang, Cho-Jui Hsieh, Xiang-Rui Wang, and Chih-Jen Lin. Liblinear: A library for large linear classification. JMLR, 2008. 16 Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick Pérez, and Matthieu Cord. Learning representations by predicting bags of visual words. In CVPR, 2020. 3 Spyros Gidaris, Andrei Bursuc, Gilles Puy, Nikos Komodakis, Matthieu Cord, and Patrick Pérez. Online bag-of-visual-words generation for unsupervised representation learning. In CVPR, 2021. 1, 3, 6, 7, 14 Priya Goyal, Piotr Dollár, Ross Girshick, Pieter Noordhuis, Lukasz Wesolowski, Aapo Kyrola, Andrew Tulloch, Yangqing Jia, and Kaiming He. Accurate, large minibatch sgd: Training imagenet in 1 hour. arXiv preprint arXiv:1706.02677, 2017. 5 Priya Goyal, Quentin Duval, Jeremy Reizenstein, Matthew Leavitt, Min Xu, Benjamin Lefaudeux, Mannat Singh, Vinicius Reis, Mathilde Caron, Piotr Bojanowski, Armand Joulin, and Ishan Misra. Vissl. https://github.com/facebookresearch/vissl, 2021. 16 Jean-Bastien Grill, Florian Str ub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In NeurIPS, 2020. 1, 3, 4, 6, 7, 8, 14, 16, 17, 20, 21 Raia Hadsell, Sumit Chopra, and Yann LeCun. Dimensionality reduction by learning an invariant mapping. In CVPR, 2006. 3 Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. I n CVPR, 2016. 5 Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask r-cnn. In ICCV, 2017. 6, 7 Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In CVPR, 2020. 1, 3, 6, 7, 16 Geoffrey Hinton, Oriol Vinyals, and Jeffrey Dean. Distilling the knowledge in a neural network. In NIPS Deep Learning and Representation Learning Workshop, 2015. 3 R Devon Hjelm, Alex Fedorov, Samuel Lavoie-Marchildon, Karan Grewal, Adam Trischler, and Yoshua Bengio. Learning deep representations by mutual information estimation and maximization. In ICLR, 2019. 1, 3 Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and detection dataset. In CVPR, 2018. 6, 7, 16 Jiabo Huang, Qi Dong andShaogang Gong, and Xiatian Zhu. Unsupervised deep learning by neighbourhood discovery. In ICML, 2019. 3 Olivier J. Hénaff, Aravind Srinivas, Jeffrey De Fauw, Ali Razavi, Carl Doersch, S. M. Ali Eslami, and Aäron van den Oord. Data-efficient image recognition with contrastive predictive coding. In ICML, 2019. 6 Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In ICML, 2015. 5 Junnan Li, Pan Zhou, Caiming Xiong, and Steven C.H. Hoi. Prototypical contrastive learning of unsupervised representations. In ICLR, 2021. 20 Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects i n context. In ECCV, 2014. 6, 7

Ilya Loshchilov and Frank Hutter. Sgdr: stochastic gradient descent with warm restarts. In *ICLR*, 2017. 5

Ishan Misra and Laurens van der Maaten. Self-supervised learning of pretext-invariant representations. In *CVPR*, 2020. 1, 3, 6, 7, 16

Karol J. Piczak. ESC: Dataset for Environmental Sound Classification. In *Proceedings of the 23rd Annual ACM Conference on Multimedia*, 2015. 18

Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In *NeurIPS*, 2015. 6, 7

Pierre H. Richemond, Jean-Bastien Grill, Florent Altché, Corentin Tallec, Florian Strub, Andrew Brock, Samuel Smith, Soham De, Razvan Pascanu, Bilal Piot, and Michal Valko. Byol works even without batch statistics. *arXiv preprint arXiv:2010.10241*, 2020. 1, 3

Yonglong Tian, Dilip Krishnan, , and Phillip Isola. Contrastive multiview coding. *arXiv preprint arXiv:1906.05849v4*, 2019. 6

Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola. What makes for good views for contrastive learning. In *NeurIPS*, 2020. 1, 6

Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. *arXiv preprint arXiv:2102.06810*, 2021. 1

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018. 3

Yuxin Wu, Alexander Kirillov, Francisco Massa, Wan-Yen Lo, and Ross Girshick. Detectron2. https://github.com/facebookresearch/detectron2, 2019. 16

Zhirong Wu, Yuanjun Xiong, Stella Yu, , and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *CVPR*, 2018. 3, 18, 20

Junyuan Xie, Ross Girshick, and Ali Farhadi. Unsupervised deep embedding for clustering analysis. In *ICML*, 2016. 3

Saining Xie, Ross Girshick, Piotr Dollár, Zhuowen Tu, and Kaiming He. Aggregated residual transformations for deep neural networks. In *CVPR*, 2017. 17

Xueting Yan, Ishan Misra, Abhinav Gupta, Deepti Ghadiyaram, and Dhruv Mahajan. Clusterfit: Improving generalization of visual representations. In *CVPR*, 2020. 3

Jianwei Yang, Devi Parikh, and Dhruv Batra. Joint unsupervised learning of deep representations and image clusters. In *CVPR*, 2016. 3

Mang Ye, Xu Zhang, Pong C Yuen, and Shih-Fu Chang. Unsupervised embedding learning via invariant and spreading instance feature. In *CVPR*, 2019. 3

Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks. *arXiv preprint arXiv:1708.03888*, 2017. 5, 17

Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. *arXiv preprint arXiv:1605.07146*, 2016. 17

Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. *arXiv preprint arxiv:2103.03230*, 2021. 1, 2, 3, 4, 5, 6, 14, 16, 20, 21

Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using places database. In *NeurIPS*, 2014. 6, 7, 16

Chengxu Zhuang, Alex Lin Zhai, and Daniel Yamins. Local aggregation for unsupervised learning of visual embeddings. In *ICCV*, 2019. 3, 18, 20

Ilya Loshchilov and Frank Hutter. Sgdr: stochastic gradient descent with warm restarts. In ICLR, 2017. 5 Ishan Misra and Laurens van der Maaten. Self-supervised learning of pretext-invariant representations. In CVPR, 2020. 1, 3, 6, 7, 16 Karol J. Piczak. ESC: Dataset for Environmental Sound Classification. In Proceedings of the 23rd Annual ACM Conference on Multimedia, 2015. 18 Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In NeurIPS, 2015. 6, 7 Pierre H. Richemond, Jean-Bastien Grill, Florent Altché, Corentin Tallec, Florian Strub, Andrew Brock, Samuel Smith, Soham De, Razvan Pascanu, Bilal Piot, and Michal Valko. Byol works even without batch statistics. arXiv preprint arXiv:2010.10241, 2020. 1, 3 Yonglong Tian, Dilip Krishnan, , and Phillip Isola. Contrastive multiview coding. arXiv preprint arXiv:1906.05849v4, 2019. 6 Yonglong Tian, Chen Sun, Ben Poole, Dilip Krishnan, Cordelia Schmid, and Phillip Isola. What makes for good views for contrastive learning. In NeurIPS, 2020. 1, 6 Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. arXiv preprint arXiv:2102.06810, 2021. 1 Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748, 2018. 3 Yuxin Wu, Alexander Kirillov, Francisco Massa, Wan-Yen Lo, and Ross Girshick. Detectron2. https://github.com/facebookresearch/detectron2, 2019. 16 Zhirong Wu, Yuanjun Xiong, Stella Yu, , and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In CVPR, 2018. 3, 18, 20 Junyuan Xie, Ross Girshick, and Ali Farhadi. Unsupervised deep embedding for clustering analysis. In ICML, 2016. 3 Saining Xie, Ross Girshick, Piotr Dollár, Zhuowen Tu, and Kaiming He. Aggregated residual transformations for deep neural networks. In CVPR, 2017. 17 Xueting Yan, Ishan Misra, Abhinav Gupta, Deepti Ghadiyaram, and Dhruv Mahajan. Clusterfit: Improving generalization of visual representations. In CVPR, 2020. 3 Jianwei Yang, Devi Parikh, and Dhruv Batra. Joint unsupervised learning of deep representations and image clusters. In CVPR, 2016. 3 Mang Ye, Xu Zhang, Pong C Yuen, and Shih-Fu Chang. Unsupervised embedding learning via invariant and spreading instance feature. In CVPR, 2019. 3 Yang You, Igor Gitman, and Boris Ginsburg. Large batch training of convolutional networks. arXiv preprint arXiv:1708.03888, 2017. 5, 17 Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. arXiv preprint arXiv:1605.07146, 2016. 17 Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow twins: Self-supervised learning via redundancy reduction. arXiv preprint arxiv:2103.03230, 2021. 1, 2, 3, 4, 5, 6, 14, 16, 20, 21 Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using places database. In NeurIPS, 2014. 6, 7, 16 Chengxu Zhuang, Alex Lin Zhai, and Daniel Yamins. Local aggregation for unsupervised learning of visual embeddings. In ICCV, 2019. 3, 18, 20

A ALGORITHM

Algorithm 1: VICReg pytorch pseudocode.

```
# f: encoder network, lambda, mu, nu: coefficients of the
   invariance, variance and covariance losses, N: batch size
   , D: dimension of the representations
# mse_loss: Mean square error loss function, off_diagonal:
   off-diagonal elements of a matrix, relu: ReLU activation
   function

for x in loader: # load a batch with N samples
   # two randomly augmented versions of x
   x_a, x_b = augment(x)

   # compute representations
   z_a = f(x_a) # N x D
   z_b = f(x_b) # N x D

   # invariance loss
   sim_loss = mse_loss(z_a, z_b)

   # variance loss
   std_z_a = torch.sqrt(z_a.var(dim=0) + 1e-04)
   std_z_b = torch.sqrt(z_b.var(dim=0) + 1e-04)
   std_loss = torch.mean(relu(1 - std_z_a)) + torch.mean(
      relu(1 - std_z_b))

   # covariance loss
   z_a = z_a - z_a.mean(dim=0)
   z_b = z_b - z_b.mean(dim=0)
   cov_z_a = (z_a.T @ z_a) / (N - 1)
   cov_z_b = (z_b.T @ z_b) / (N - 1)
   cov_loss = off_diagonal(cov_z_a).pow_(2).sum() / D
              + off_diagonal(cov_z_b).pow_(2).sum() / D

   # loss
   loss = lambda * sim_loss + mu * std_loss + nu * cov_loss

   # optimization step
   loss.backward()
   optimizer.step()
```


A. 算法

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   cov_loss = off_diagonal(cov_z_a).pow_(2).sum() / D
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   loss.backward()
   optimizer.step()
```

B RELATION TO OTHER SELF-SUPERVISED METHODS

We compare here VICReg with other methods in terms of methodology, and we discuss the mechanisms used by these methods to avoid collapse and to learn representations, and how they relate to VICReg. We synthesize and illustrate the differences between these methods in Figure 2.

Relation to Barlow Twins Zbontar et al. (2021). VICReg uses the same decorrelation mechanism as Barlow Twins, which consists in penalizing the off-diagonal terms of a covariance matrix computed on the embeddings. However, Barlow Twins uses the cross-correlation matrix where each entry in the matrix is a cross-correlation between two vectors z^i and z'^j, from the two branches of the siamese architecture. Instead of using cross-correlations, we simply use the covariance matrix of each branch individually, and the variance term of VICReg allows us to get rid of standardization. Indeed, Barlow Twins forces the correlations between pairs of vectors z^i and z'^i from the same dimension i to be 1. Without normalization, this target value of 1 becomes arbitrary and the vectors take values in a wider range. Moreover, there is an undesirable phenomenon happening in Barlow Twins, the embeddings before standardization can shrink and become constant to numerical precision, which could cause numerical instabilities. In practice, this is solved by adding a constant scalar in the denominator of standardization of the embeddings. Without normalization, VICReg naturally avoids this edge case.

Relation to W-MSE Ermolov et al. (2021). The whitening operation of W-MSE consists in computing the inverse covariance matrix of the embeddings and use its square root as a whitening operator on the embeddings. Using this operator has two downsides. First, matrix inversion is a very costly and potentially unstable operation. VICReg does not need to inverse the covariance matrix. Second, as mentioned in Ermolov et al. (2021) the whitening operator is constructed over several consecutive iteration batches and therefore might have a high variance, which biases the estimation of the mean-squared error. This issue is overcome in practice by a batch slicing strategy, where the whitening operator is computed over randomly constructed sub-batches. VICReg does not apply any operator on the embeddings, but instead regularizes the variance and covariance of the embeddings using an additional constraint.

Relation to BYOL and SimSiam Grill et al. (2020); Chen & He (2020). The core components that avoid collapse in BYOL and SimSiam are the average moving weights and the stop-gradient operation on one side of their asymmetric architecture, which play the role of the repulsive term used in other methods. Our experiments in Appendix D.8 show that in addition to preventing collapse, these components also have a decorrelation effect. In addition, we have conducted the following experiment: We compute the correlation matrix of the final representations obtained with SimSiam, BYOL, VICReg and VICReg without covariance regularization. We measure the average correlation coefficient and observe that this coefficient is much smaller for SimSiam, BYOL and VICReg, compared to VICReg without covariance regularization. We observe in Figure 5 that even without covariance regularization, SimSiam and BYOL naturally minimize the average correlation coefficient of the representations. VICReg replaces the moving average weights and the stop-gradient operation, which are architectural trick that require some dependency between the branches, by an explicit constraint on the variance and the covariance of both embeddings separately, which achieves the same goal of decorrelating the representations and avoiding collapse, while being clearer, more interpretable, and working with independent branches.

Relation to SimCLR, SwAV and OBoW Caron et al. (2020); Chen et al. (2020a); Gidaris et al. (2021). Contrastive and clustering based self-supervised algorithms rely on direct comparisons between elements of negative pairs. In the case of SimCLR, the negative pairs involve embeddings mined from the current batch, and large batch sizes are required. Despite the fact that SwAV computes clusters using elements in the current batch, it does not seem to have the same dependency on batch size. However, it still requires a lot of prototype vectors for negative comparisons between embeddings and codes. VICReg eliminates the negative comparisons and replace them by an explicit constraint on the variance of the embeddings, which efficiently plays the role of a negative term between the vectors. SwAV can also be interpreted as a distillation method, where a teacher network produces quantized vectors, used as target for a student network. Ensuring an equal partition of the quantized vectors in different bins or clusters effectively prevents collapse. OBOW can also be interpreted under the same framework. The embeddings are bag-of-words over a vocabulary of visual features, and collapse is avoided by the underlying quantization operation.

B. 与其他自监督方法的关系

这里从方法角度比较 VICReg 与其他方法，讨论它们防坍塌、学习表征的机制及与 VICReg 的联系。图2汇总并图示差异。

与 Barlow Twins 的关系（Zbontar et al., 2021） VICReg 使用与 Barlow Twins 相同的去相关机制，即惩罚嵌入协方差矩阵的非对角项。但 Barlow Twins 使用交叉相关矩阵，每项是孪生结构两侧向量 z^i 和 z'^j 的交叉相关。我们仅分别使用各分支协方差矩阵，并通过方差项取消标准化。Barlow Twins 将同一维度 i 的向量对 z^i 、 z'^i 的相关性迫近1；没有归一化，目标1就显得任意，向量取值范围也更宽。此外，Barlow Twins 有不理想现象：标准化前的嵌入可能缩小，并在数值精度范围内成为常量，导致数值不稳定。实际中，通过在嵌入标准化分母中加入常量解决。无需归一化的 VICReg 自然避免这一边界情况。

与 W-MSE 的关系（Ermolov et al., 2021） W-MSE 白化先计算嵌入协方差矩阵的逆，再以其平方根作为白化算子。存在两项缺点：首先，矩阵求逆计算昂贵且可能不稳定，VICReg 无需协方差求逆。其次，如原文所述，白化算子由连续若干迭代批次构建，可能方差较大，使均方误差估计有偏。实际中通过批次切片，在随机构建的子批次上计算白化算子解决。VICReg 不对嵌入施加算子，而用额外约束正则化方差和协方差。

与 BYOL 和 SimSiam 的关系（Grill et al., 2020; Chen & He, 2020） 两者核心防坍塌组件是移动平均权重及非对称结构一侧的停止梯度，发挥其他方法中排斥项的作用。附录 D.8表明，它们除了防坍塌，还具有去相关效果。我们进一步计算 SimSiam、BYOL、VICReg 及无协方差正则的 VICReg 最终表征的相关矩阵。平均相关系数显示，前三者明显小于无协方差正则的 VICReg。图5表明，即使不加协方差正则，SimSiam 与 BYOL 也会自然减小表征的平均相关系数。VICReg 用两组嵌入各自的显式方差和协方差约束，取代要求分支依赖的移动平均与停止梯度结构技巧，同样实现表征去相关和防坍塌；同时更清晰、更可解释，并允许分支独立。

与 SimCLR、SwAV、OBoW 的关系（Caron et al., 2020; Chen et al., 2020a; Gidaris et al., 2021） 对比与聚类自监督算法依赖对负样本对元素的直接比较。SimCLR 从当前批次挖掘负样本嵌入，需要大批量。SwAV 也用当前批次聚类，但似乎没有同样的批量依赖；不过它仍需大量原型向量，对嵌入与编码进行负向比较。VICReg 取消负向比较，改用显式嵌入方差约束，高效地发挥向量间负项的作用。SwAV 也可理解为蒸馏：教师产生量化向量作为学生目标，通过均匀分配到不同分箱或聚类来防坍塌。OBoW 也可用同一框架理解，其嵌入是视觉特征词汇表上的词袋，通过底层量化操作防坍塌。

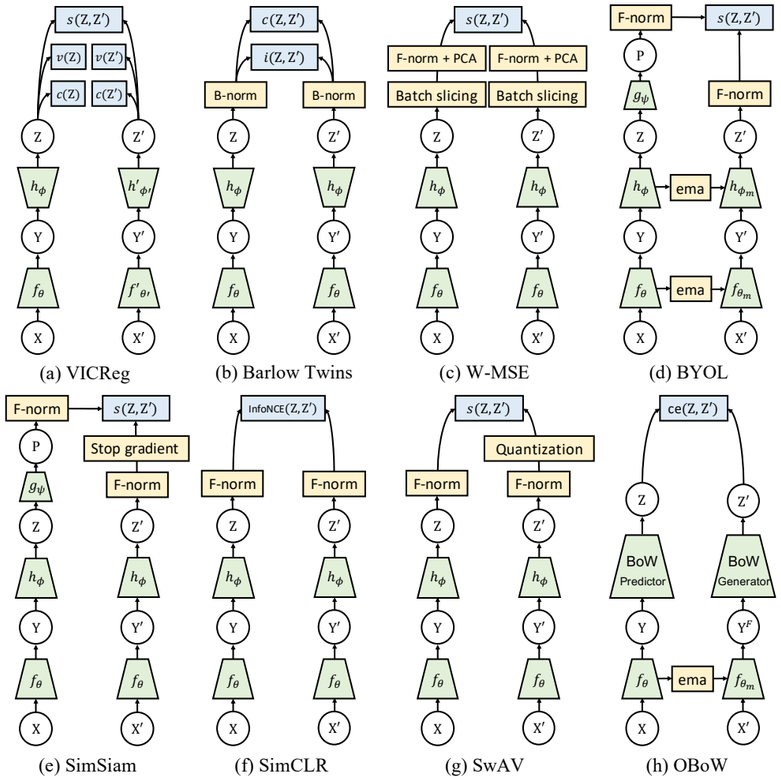


Figure 2: **Conceptual comparison between different self-supervised methods.** The inputs X and X' are fed to an encoder f with weights θ. The representations Y and Y' are further processed by a network h with weights ψ. h can be a projector (narrowing trapeze) that reduces the dimensionality of the representations, or an expander (widening trapeze) that increases their dimensionality. A criterion is finally applied on the embeddings Z and Z'. VICReg (a) works when both branches have encoders f and f' with different architectures and sets of weights θ and θ'. Each branch's variance and covariance are regularized by regularizers v and c, and the distance between both branches is minimized with a mean-squared error loss s. Barlow Twins (b) uses a loss c to decorrelate pairs of different dimensions in the batch-wise normalized (B-Norm) embeddings, and learns invariance with a loss i that makes similar dimensions highly correlated. W-MSE (c) uses a batch slicing operation that shuffles batches into small sub-batches, and apply PCA as a whitening operation on the feature-wise normalized (F-Norm) embeddings of each sub-batch. BYOL (d) has an asymmetric architecture where the weights θ_m of one encoder are an exponential moving average (ema) of the other encoder's weights θ. A predictor g with weights ψ is used in the branch with learnable weights. SimSiam (e) uses a predictor on one branch and a stop-gradient operation (sg) on the other one. SimCLR (f) uses the InfoNCE contrastive loss where all the feature-wise normalized embeddings are compared between them inside a batch. Samples from distorted versions of the same input are brought close to each other, while other samples are pushed away. SwAV (g) quantizes the feature-wise normalized embeddings of a branch and use it as target for the other one. OBoW (h) uses bag-of-words (BoW) representations and a cross-entropy loss to compare the BoW generated by a teacher network from the feature maps Y^F of the encoder, to the BoW predicted by a student network. Green blocks: parametric functions; yellow boxes: non-parametric functions; blue boxes: objective functions.

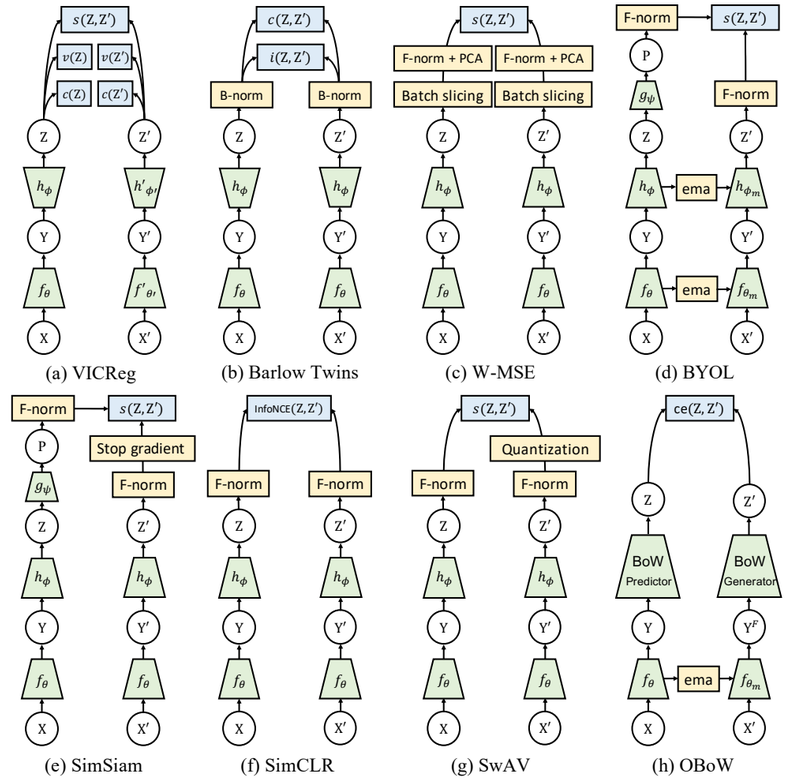


图2：不同自监督方法的概念比较。输入 X 和 X' 送入权重为 θ 的编码器 f。表征 Y、Y' 再由权重为 ψ 的网络 h 处理。h 可为降低表征维度的投影器（变窄梯形），或提高维度的扩展器（变宽梯形）。最后在嵌入 Z 和 Z' 上计算损失。VICReg（a）允许两分支编码器 f、f' 采用不同结构与权重 θ、θ'。每个分支由 v、c 正则化方差与协方差，并用均方误差 s 最小化两分支距离。Barlow Twins（b）用损失 c 使沿批量归一化（B-Norm）嵌入的不同维度对去相关，用损失 i 使对应维度高度相关，从而学习不变性。W-MSE（c）通过批次切片，将打乱的批次分成小批次，并对各小批次沿特征归一化（F-Norm）的嵌入做 PCA 白化。BYOL（d）采用非对称结构，一个编码器的权重 θ_m 是另一编码器权重 θ 的指数移动平均（ema），可训练分支上有权重为 ψ 的预测器 g。SimSiam（e）一侧使用预测器，另一侧停止梯度（sg）。SimCLR（f）采用 InfoNCE 对比损失，在批内比较全部按特征归一化的嵌入；同一输入不同扰动的样本被拉近，其他样本被推远。SwAV（g）量化一侧按特征归一化的嵌入，作为另一侧的目标。OBoW（h）使用词袋（BoW）表征与交叉熵，比较教师网络从编码器特征图 Y^F 生成的词袋，与学生网络预测的词袋。绿色：参数化函数；黄色：非参数函数；蓝色：目标函数。

C ADDITIONAL IMPLEMENTATION DETAILS

C.1 DATA AUGMENTATION

We follow the image augmentation protocol first introduced in SimCLR Chen et al. (2020a) and now commonly used by similar approaches based on siamese networks Caron et al. (2020); Grill et al. (2020); Chen & He (2020); Zbontar et al. (2021). Two random crops from the input image are sampled and resized to 224×224, followed by random horizontal flip, color jittering of brightness, contrast, saturation and hue, Gaussian blur and random grayscale. Each crop is normalized in each color channel using the ImageNet mean and standard deviation pixel values. In more details, the exact set of augmentations is based on BYOL Grill et al. (2020) data augmentation pipeline but is symmetrised. The following operations are performed sequentially to produce each view:

- Random cropping with an area uniformly sampled with size ratio between 0.08 to 1.0, followed by resizing to size 224×224. `RandomResizedCrop(224, scale=(0.08, 0.1))` in PyTorch.
- Random horizontal flip with probability 0.5.
- Color jittering of brightness, contrast, saturation and hue, with probability 0.8. `ColorJitter(0.4, 0.4, 0.2, 0.1)` in PyTorch.
- Grayscale with probability 0.2.
- Gaussian blur with probability 0.5 and kernel size 23.
- Solarization with probability 0.1.
- color normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

C.2 IMAGENET EVALUATION

Linear evaluation. We follow standard procedure and train a linear classifier on top of the frozen representations of a ResNet-50 pretrained with VICReg. We use the SGD optimizer with a learning rate of 0.02, a weight decay of 10^{-6}, a batch size of 256, and train for 100 epochs. The learning rate follows a cosine decay. The training data augmentation pipeline is composed of random cropping and resize of ratio 0.2 to 1.0 with size 224×224, and random horizontal flips. During evaluation the validation images are simply center cropped and resized to 224×224.

Semi-supervised evaluation. We train a linear classifier and fine-tune the representations using 1 and 10% of the labels. We use the SGD optimizer with no weight decay and a batch size of 256, and train for 20 epochs. We perform a grid search on the values of the encoder and linear head learning rates. In the 10% of labels case, we use a learning rate of 0.01 for the encoder and 0.1 for the linear head. In the 1% of labels case we use 0.03 for the encoder and 0.08 for the linear head. The two learning rates follow a cosine decay schedule. The training data and validation augmentation pipelines are identical to the linear evaluation data augmentation pipelines.

C.3 TRANSFER LEARNING

We use the VISSL library Goyal et al. (2021) for linear classification tasks and the detectron2 library Wu et al. (2019) for object detection and segmentation tasks.

Linear classification. We follow standard protocols Misra & Maaten (2020); Caron et al. (2020); Zbontar et al. (2021) and train linear models on top of the frozen representations. For VOC07 Everingham et al. (2010), we train a linear SVM with LIBLINEAR Fan et al. (2008). The images are center cropped and resized to 224×224, and the C values are computed with cross-validation. For Places205 Zhou et al. (2014) we use SGD with a learning rate of 0.003, a weight decay of 0.0001, a momentum of 0.9 and a batch size of 256, for 28 epochs. The learning rate is divided by 10 at epochs 4, 8 and 12. For Inaturalist2018 Horn et al. (2018), we use SGD with a learning rate of 0.005, a weight decay of 0.0001, a momentum of 0.9 and a batch size of 256, for 84 epochs. The learning rate is divided by 10 at epochs 24, 48 and 72.

Object detection and instance segmentation. Following the setup of He et al. (2020); Zbontar et al. (2021), we use the `trainval` split of VOC07+12 with 16K images for training and a Faster

C. 补充实现细节

C.1. 数据增强

我们遵循 SimCLR（Chen et al., 2020a）首创、现广泛用于孪生网络方法的图像增强流程（Caron et al., 2020; Grill et al., 2020; Chen & He, 2020; Zbontar et al., 2021）：抽取两个随机裁剪并缩放到 224×224，随后随机水平翻转、亮度／对比度／饱和度／色调抖动、高斯模糊及随机灰度化。每个裁剪按通道使用 ImageNet 像素均值与标准差归一化。具体增强集合基于 BYOL（Grill et al., 2020），但已对称化。各视图依次执行：

- 随机裁剪，面积比例从0.08至1.0均匀抽取，再缩放到 224×224。PyTorch 写法：RandomResizedCrop(224, scale=(0.08, 0.1))。
- 随机水平翻转，概率0.5。
- 以0.8概率对亮度、对比度、饱和度和色调进行颜色抖动。PyTorch：ColorJitter(0.4, 0.4, 0.2, 0.1)。
- 灰度化，概率0.2。
- 高斯模糊，概率0.5，核大小23。
- 曝光反转，概率0.1。
- 颜色归一化，均值（0.485, 0.456, 0.406），标准差（0.229, 0.224, 0.225）。

C.2. ImageNet 评估

线性评估 遵循标准做法，在 VICReg 预训练 ResNet-50 的冻结表征上训练线性分类器。SGD，学习率0.02，权重衰减 10^{-6}，批量256，训练100轮，学习率余弦衰减。训练增强为面积比例0.2至1.0的随机裁剪、缩放到 224×224 及随机水平翻转。评估时仅将验证图像中心裁剪并缩放到 224×224。

半监督评估 使用1%及10%标注训练线性分类器并微调表征。SGD，无权重衰减，批量256，训练20轮；对编码器和线性头学习率进行网格搜索。10%标注时，编码器0.01、线性头0.1；1%标注时，编码器0.03、线性头0.08。两者均采用余弦衰减。训练与验证增强流程和线性评估相同。

C.3. 迁移学习

线性分类使用 VISSL（Goyal et al., 2021），检测和分割使用 detectron2（Wu et al., 2019）。

线性分类 遵循标准流程（Misra & Maaten, 2020; Caron et al., 2020; Zbontar et al., 2021），在冻结表征上训练线性模型。VOC07（Everingham et al., 2010）使用 LIBLINEAR（Fan et al., 2008）训练线性 SVM；图像中心裁剪并缩放到 224×224，C 值通过交叉验证确定。Places205（Zhou et al., 2014）用 SGD 训练28轮，学习率0.003，权重衰减0.0001，动量0.9，批量256；第4、8、12轮学习率各除以10。iNaturalist2018（Horn et al., 2018）用 SGD 训练84轮，学习率0.005，权重衰减0.0001，动量0.9，批量256；第24、48、72轮学习率各除以10。

目标检测与实例分割 遵循 He et al.（2020）及 Zbontar et al.（2021），用 VOC07+12 trainval 的16K张图像训练，采用 Faster

R-CNN C-4 backbone for 24K iterations with a batch size of 16. The backbone is initialized with our pretrained ResNet-50 backbone. We use a learning rate of 0.1, divided by 10 at iteration 18K and 22K, a linear warmup with slope of 0.333 for 1000 iterations, and a region proposal network loss weight of 0.2. For COCO we use Mask R-CNN FPN backbone for 90K iterations with a batch size of 16, a learning rate of 0.04, divided by 10 at iteration 60K and 80K and with 50 warmup iterations.

C.4 Analysis

We give here implementation details on the results of Table 4 with BYOL and SimSiam, as well as the default setup for VICReg with 100 epochs of pretraining, used in all our ablations included in Appendix D. For both BYOL and SimSiam experiments, the variance criterion has coefficient $\mu = 1$ and the covariance criterion has coefficient $\nu = 0.01$, the data augmentation pipeline and the architectures of the expander and predictor exactly follow the pipeline and architectures described in their paper. The linear evaluation setup of each methods follows closely the setup described in the original papers.

BYOL setup. We use our own BYOL implementation in PyTorch, which outperforms the original implementation for 100 epochs of pretraining (69.3% accuracy on the linear evaluation protocol against 66.5% for the original implementation) and matches its performance for 1000 epochs of pretraining. We use the LARS optimizer You et al. (2017), with a learning rate of $base_lr * batch_size/256$ where $base_lr$ = 0.45, and $batch_size$ = 4096, a weight decay of 10^{-6}, an eta value of 0.001 and a momentum of 0.9, for 100 epoch of pretraining with 10 epochs of warmup. The learning rate follows a cosine decay schedule. The initial value of the exponential moving average factor is 0.99 and follows a cosine decay schedule.

SimSiam setup. We use our own implementation of SimSiam, which reproduces exactly the performance reported in the paper Chen & He (2020). We use SGD with a learning rate of $base_lr * batch_size/256$ where $base_lr$ = 0.05, $batch_size$ = 2048, with a weight decay of 0.0001 and a momentum of 0.9 for 100 epochs of pretraining and 10 epochs of warmup. The learning rate of the encoder and the expander follow a cosine decay schedule while the learning rate of the predictor is kept fixed.

VICReg setup. The setting of VICReg's experiments is identical to the setting described in section 4.2, except that the number of pretraining epochs is 100 and the base learning rate is 0.3. The base learning rates used for the batch size study are 0.8, 0.5 and 0.4 for batch size 128, 256 and 512 respectively, and 0.3 for all other batch sizes. When a predictor is used, it has a similar architecture as the expander described in section 4.2, but with 2 layers instead of 3, which gives better results in practice.

D Additional results

D.1 Other ResNet architectures

Table 9 reports the performance of VICReg on linear classification with large ResNet architectures. We focus on the wider family of ResNet Zagoruyko & Komodakis (2016) and aggregated ResNet Xie et al. (2017), and we consider two ways of widening a standard ResNet. First, we follow standard practice in recent self-supervised learning work Caron et al. (2020); Grill et al. (2020); Chen et al. (2020a) and multiple by 2 or 4 the number of filters in every convolutional layer, which also has the effect of multiplying the dimensionality of the representations. Second, as originally proposed in Zagoruyko & Komodakis (2016), we only multiply the number of filters in the bottleneck layers, which does not increases the dimensionality of the representations. We call this architecture Narrow ResNet (with prefix N- in Table 9). The main observation we make is the dependency of VICReg on the dimensionality of the representation. Using the narrow architecture, the performance of VICReg, jumps from 73.2% top-1 accuracy on linear classification with a ResNet-50, to 74.7% with Narrow ResNet-50 (x2), which is a 1.5% improvement and 76.0% with Narrow ResNet-50 (x4), which is a 2.8% improvement. We observe a similar trend going from ResNet-50 to ResNet-50 (x2), which is a 2.3% improvement but the performance completely saturates with ResNet-50 (x4), which is a 0.1% improvement over ResNet-50 (x2). Table 10 reports the performance of VICReg on semi-supervised classification with large ResNet architectures. VICReg combined with a ResNet-50 (x2) outperforms the current state-of-the-art methods BYOL and SimCLR, using this encoder architecture. Our largest

R-CNN C-4 主干训练24K次迭代，批量16，以预训练 ResNet-50 初始化。学习率0.1，第18K及22K次各除以10；预热1000次，线性斜率0.333；区域提议网络损失权重0.2。COCO 使用 Mask R-CNN FPN 主干，训练90K次，批量16，学习率0.04；第60K及80K次各除以10，预热50次。

C.4. 分析

这里给出表4 BYOL 和 SimSiam 的实现细节，以及附录D全部消融采用的100轮 VICReg 默认设置。BYOL、SimSiam 的方差项系数为 $\mu = 1$，协方差项系数为 $\nu = 0.01$；增强流程以及扩展器、预测器结构完全遵循原论文。各方法线性评估设置也严格遵循原文。

BYOL 设置 我们使用自行实现的 PyTorch BYOL，预训练100轮时优于原实现（线性准确率69.3%，原实现66.5%），1000轮时与其相当。使用 LARS（You et al., 2017），学习率 $base_lr * batch_size/256$，其中 $base_lr$ = 0.45，且 $batch_size$ = 4096；权重衰减 10^{-6}，eta 为0.001，动量0.9。预训练100轮，预热10轮，学习率采用余弦衰减。指数移动平均因子的初始值0.99，采用余弦衰减调度。

SimSiam 设置 我们自行实现 SimSiam，准确复现 Chen & He（2020）结果。SGD 学习率 $base_lr * batch_size/256$，其中 $base_lr$ = 0.05、$batch_size$ = 2048；权重衰减 0.0001，动量0.9。预训练100轮，预热10轮。编码器和扩展器学习率余弦衰减，预测器学习率固定。

VICReg 设置 与第4.2节相同，但预训练100轮，基础学习率0.3。批量研究中，128、256、512分别采用0.8、0.5、0.4，其余批量均0.3。预测器若存在，结构类似第4.2节扩展器，但由三层改为两层，实际效果更好。

D. 补充结果

D.1. 其他 ResNet 结构

表9报告 VICReg 在大型 ResNet 结构上的线性分类性能。我们关注更宽的 ResNet（Zagoruyko & Komodakis, 2016）及聚合式 ResNet（Xie et al., 2017），考虑两种加宽方式。首先，遵循近期自监督工作的常用做法（Caron et al., 2020; Grill et al., 2020; Chen et al., 2020a），将每个卷积层的滤波器数乘以2或4，同时提高表征维度。其次，遵循 Zagoruyko & Komodakis（2016）的原始建议，仅增加瓶颈层滤波器，不提高表征维度；称为窄型 ResNet（表9以 N- 为前缀）。主要发现是 VICReg 对表征维度的依赖。采用窄型结构时，top-1 线性分类准确率由 ResNet-50 的73.2%提升到窄型 ResNet-50（×2）的74.7%，提高1.5个百分点；窄型 ResNet-50（×4）达到76.0%，提高2.8个百分点。从 ResNet-50 到 ResNet-50（×2）也提升2.3个百分点，但 ResNet-50（×4）完全饱和，仅比×2高0.1个百分点。表10报告大型 ResNet 半监督分类性能。VICReg 配合 ResNet-50（×2），优于采用相同编码器的当前最优 BYOL 与 SimCLR。我们最大的

Table 6: **Evaluation on ESC-50.** Evaluation of the representations obtained with a ResNet-18 backbone pretrained with VICReg on ESC-50 Piczak (2015) by processing jointly a raw audio time-series and its corresponding time-frequency representation. The supervised baseline corresponds to a ResNet-18 trained on the time-frequency representation in a supervised way. We report Top-1 accuracy on the validation set (in %).

Method	Top-1
Supervised baseline	72.7
Barlow Twins	75.4
VICReg	78.4

model ResNet-200 (x2) performs lower than BYOL when 1% of the labels are used but is on par with 10% of the labels. These results demonstrate the capabilities of VICReg to scale up when large architectures are used.

D.2 Pretraining and evaluation on ESC-50 audio classification

We demonstrate the ability of VICReg to function in a setting where the branches have different architectures by pretraining on the ESC-50 audio dataset Piczak (2015), which is an environmental sound classification dataset with 50 classes. We jointly embedded a raw audio time-series representation on one branch, with its corresponding time-frequency representation on the other branch. We use the standard split of ESC-50 Piczak (2015), composed of 1600 training audio samples and 400 validation sample. The raw audio encoder is a 1-dimensional ResNet-18 with output dimension 384. The time-frequency image representation is the mel spectrogram with 1 channel of the raw audio, that we normalize between 0 and 1, and that is processed be a ResNet-18 with output dimension of 512. We use the AdamW optimizer with learning rate 0.0005 for 100 epochs of pretraining.

Table 6 reports the performance of a linear classifier trained one the frozen representations obtained with VICReg and Barlow Twins to a simple supervised baseline where we train a ResNet-18 on the time-frequency representation in a supervised way. VICReg performs better by 5.7% than our supervised baseline, and better by 3.0% than Barlow Twins. We give more details in Appendix **??**. Current best approaches that report around 95% accuracy on this task uses tricks such as heavy data augmentation or pretraining on larger audio and video datasets. With this experiment, our purpose is not to push the state of the art on ESC-50, but merely to demonstrate the applicability of VICReg to settings with multiple architectures and input modalities.

D.3 K-nearest-neighbors

Following recent protocols Caron et al. (2020); Wu et al. (2018); Zhuang et al. (2019), we evaluate the learnt representations using K-nearest-neighbors classifiers built on the training set of ImageNet and evaluated on the validation set of ImageNet. We report the results with K=20 and K=200 in Table 11. VICReg performs slightly lower than other methods in the 20-NN case but remains competitive in the 200-NN case. These results with K-NN classifiers demonstrate the potential applicability of VICReg to downstream tasks based on nearest neighbors search, such as content retrieval in images or videos.

D.4 Loss function coefficients.

Table 7 reports the performance for various values of the loss term coefficients in Eq. (6). Without variance regularization the representations immediately collapse to a single vector and the covariance term, which has no repulsive effect preventing collapse, has no impact. The invariance term is absolutely necessary and without it the network can not learn any good representations. By simply using the invariance term and variance regularization, which is a very simple baseline, VICReg still reaches an accuracy of 57.5%. These results show that variance and covariance regularizations have complementary effects, and that both are required.

On ImageNet, we choose the final coefficients the following way. First, we have empirically found that using very different values for λ and μ, or taking $\lambda = \mu$ with $\nu > \mu$ leads to unstable training. On the other hand taking $\lambda = \mu$ and picking $\nu < \mu$ leads to stable convergence, with the exact value picked for mu having very limited influence on the final linear classification accuracy. We have found

表6：ESC-50 评估。VICReg 在 ESC-50（Piczak, 2015）上联合处理原始音频时间序列与对应时频表征，预训练 ResNet-18。监督基线是在时频表征上监督训练的 ResNet-18。报告验证集 top-1 准确率（%）。

方法	Top-1
监督基线	72.7
Barlow Twins	75.4
VICReg	78.4

模型 ResNet-200（×2）在1%标注时低于 BYOL，10%时相当。这些结果展示 VICReg 能扩展到大型结构。

D.2. ESC-50 音频分类预训练与评估

我们在 ESC-50 音频数据集（Piczak, 2015）上预训练，验证 VICReg 能处理分支结构不同的情形。ESC-50 是50类环境声音分类数据集。一个分支嵌入原始音频时间序列，另一分支嵌入其对应时频表征。采用标准划分：1600个训练音频样本、400个验证样本。原始音频编码器为一维 ResNet-18，输出384维。时频图像为单通道梅尔频谱，归一化到0至1，经 ResNet-18 处理，输出512维。用 AdamW 优化器，学习率0.0005，预训练100轮。

表6将 VICReg、Barlow Twins 冻结表征上的线性分类器，与直接在时频表征上监督训练 ResNet-18 的简单基线比较。VICReg 比监督基线高5.7个百分点，比 Barlow Twins 高 3.0个百分点。更多细节见附录??。当前在该任务上达到约95%准确率的方法，采用大量增强或在更大音视频数据集上预训练等技巧。本实验旨在验证 VICReg 对多结构、多输入模态的适用性，而非刷新 ESC-50 最佳水平。

D.3. K 近邻分类

遵循近期流程（Caron et al., 2020; Wu et al., 2018; Zhuang et al., 2019），在 ImageNet 训练集上构建 K 近邻分类器，在验证集评估表征。表11报告 K=20 和 K=200 的结果。VICReg 在20近邻时略低于其他方法，200近邻时仍有竞争力。这说明它可能适用于基于近邻搜索的下游任务，如图像或视频内容检索。

D.4. 损失函数系数

表7报告式6中不同损失系数的性能。不加方差正则时，表征立即坍塌成单一向量；协方差项没有防坍塌排斥效果，因此不起作用。不变性项必不可少，否则网络无法学习有用表征。仅使用不变性项及方差正则这一非常简单的基线，VICReg 仍达到57.5%。说明方差与协方差正则作用互补，两者都需要。

ImageNet 最终系数选择如下。首先，实验发现 λ 与 μ 差异很大，或在 $\nu > \mu$ 时采用 $\lambda = \mu$，会使训练不稳定。另一方面，采用 $\lambda = \mu$ 并选择 $\nu < \mu$，会稳定收敛，mu 的具体取值对最终线性准确率影响很小。我们发现

Table 7: **Impact of variance-covariance regularization.** Inv: a invariance loss is used, $\lambda > 0$, Var: variance regularization, $\mu > 0$, Cov: covariance regularization, $\nu > 0$, in Eq. (6).

Method	λ	μ	ν	Top-1
Inv	1	0	0	collapse
Inv + Cov	25	0	1	collapse
Inv + Cov	0	25	1	collapse
Inv + Var	1	1	0	57.5
Inv + Var + Cov (VICReg)	1	1	1	collapse
	1	10	1	collapse
	10	1	1	collapse
	5	5	1	68.1
	10	10	1	68.2
	25	25	1	68.6
	50	50	1	68.3

Table 8: **Impact of normalization.** Std: variables are centered and divided by their standard deviation over the batch. This is applied or not to the embedding and the expander hidden layers. l_2: the embedding vectors are l_2-normalized.

Representation	Embedding	Top-1
Std	None	68.6
Std	Std	68.4
None	Std	67.4
Std	None	67.2
None	l_2	65.1

that setting $lambda = mu = 25$ and $nu = 1$ works best (by a small margin) for Imagenet but we have also obtained excellent results on MNIST and Cifar-10 and 100 using these exact same values. We could easily have tuned these parameters by cross-validation on the validation sets of these two smaller datasets.

D.5 NORMALIZATIONS

VICReg is the first self-supervised method for joint-embedding architectures we are aware of that does not require normalization. Contrary to SimSiam, W-MSE, SwAV and BYOL, and others, the embedding vectors are not projected on the unit sphere. Contrary to Barlow Twins, they are not standardized (equivalent to batch normalization without the adaptive parameters). Table 8 shows that the best settings do not involve any normalization of the embeddings, whether it is batch-wise or feature-wise (as in l_2 normalization). Whenever the embeddings are standardized (lines 3 and 5 in the table) the covariance matrix of Eq. (3) becomes the normalized auto-correlation matrix with coefficients between -1 and 1. This hurts the accuracy by 0.2%. We observe that when unconstrained, the coefficients in the covariance matrix take values in a wider range, which seems to facilitate the training process. Standardization is still an important component that helps stabilize the training when used in the hidden layers of the expander, and the performance drops by 1.2% when it is removed. Projecting the embeddings on the unit sphere implicitly constrains their standard deviation along the batch dimension to be $1/\sqrt{d}$, where d is the dimension of the vectors. We change the invariance term of Eq. (5) to be the mean square error between l_2-normalized vectors, and the target γ in the variance term of Eq. (1) is set to $1/\sqrt{d}$ instead of 1, forcing the standard deviation to get closer to $1/\sqrt{d}$, and the vectors to be spread out on the unit sphere. This puts a lot more constraints on the network and the performance drops by 3.5%.

表7：方差—协方差正则的影响。Inv：不变性损失，$\lambda > 0$；Var：方差正则，$\mu > 0$；Cov：协方差正则，$\nu > 0$，见式6。

方法	λ	μ	ν	Top-1
Inv	1	0	0	坍塌
不变性 + 协方差	25	0	1	坍塌
不变性 + 协方差	0	25	1	坍塌
不变性 + 方差	1	1	0	57.5
不变性 + 方差 + 协方差（VICReg）	1	1	1	坍塌
	1	10	1	坍塌
	10	1	1	坍塌
	5	5	1	68.1
	10	10	1	68.2
	25	25	1	68.6
	50	50	1	68.3

表8：归一化的影响。Std：变量中心化并除以批内标准差，可用于嵌入和扩展器隐藏层。l_2：嵌入向量进行 l_2 归一化。

表征	嵌入	Top-1
Std	None	68.6
Std	Std	68.4
None	Std	67.4
Std	None	67.2
None	l_2	65.1

设定 $lambda = mu = 25$、$nu = 1$ 在 ImageNet 上略微最佳；相同取值也在 MNIST、CIFAR-10 和 CIFAR-100 上获得优秀结果。对于这些较小数据集，我们本可以很容易地在验证集交叉验证调参。

D.5. 归一化

据我们所知，VICReg 是首个不需要归一化的联合嵌入自监督方法。不同于 SimSiam、W-MSE、SwAV、BYOL 等，嵌入不投影到单位球面；不同于 Barlow Twins，也不进行标准化，即不做无仿射参数的批归一化。表8显示，最佳设置不包含任何批量或特征方向的嵌入归一化（如 l_2 归一化）。嵌入标准化时（表中第3、5行），式3的协方差矩阵变为归一化自相关矩阵，系数在−1与1之间，准确率下降0.2个百分点。不加约束时，协方差系数取值范围更宽，似乎有助训练。扩展器隐藏层的标准化仍有助稳定训练，移除后性能下降1.2个百分点。将嵌入投影到单位球面，会隐式约束沿批量维度的标准差为 $1/\sqrt{d}$，其中 d 是向量维度。我们将式5不变性项改为 l_2 归一化向量之间的均方误差，并将式1方差项的目标 γ 从1改为 $1/\sqrt{d}$，迫使标准差接近 $1/\sqrt{d}$，使向量在单位球面上分散。这对网络施加更多约束，性能下降3.5个百分点。

Table 9: **Linear classification with large architectures.** Top-1 accuracy comparison between different methods using various encoder architectures. For all VICReg results, the output dimensionality of the expander is 8192. N-R stands for Narrow ResNet, where only the bottleneck convolutional layers are widen.

Method	Arch.	Param.	Repr.	Top-1	Top-5
SimCLR Chen et al. (2020a)	R50 (x2)	93M	4096	74.2	92.0
	R50 (x4)	375M	8192	76.5	93.2
SwAV Caron et al. (2020)	R50 (x2)	93M	4096	77.3	-
	R50 (x4)	375M	8192	77.9	-
	R50 (x5)	586M	10240	78.5	-
BYOL Grill et al. (2020)	R50 (x2)	93M	4096	77.4	93.6
	R50 (x4)	375M	8192	78.6	94.2
	R200 (x2)	250M	4096	79.6	94.8
VICReg (ours)	N-R50 (x2)	66M	2048	74.7	91.9
	N-R50 (x4)	221M	2048	76.0	92.4
	R50 (x2)	93M	4096	75.5	92.1
	R50 (x4)	375M	8192	75.6	92.2
	RNXT101-32-16	191M	2048	76.1	92.3
	R200 (x2)	250M	4096	77.3	93.3

Table 10: **Semi-supervised classification with large architectures.** Top-1 accuracy comparison between different methods using various encoder architectures. For all VICReg results, the output dimensionality of the expander is 8192.

Method	Arch.	Param.	Repr.	Top-1		Top-5	
				1%	10%	1%	10 %
SimCLR Chen et al. (2020a)	R50 (x2)	93M	4096	58.5	71.7	83.0	91.2
	R50 (x4)	375M	8192	63.0	74.4	85.8	92.6
BYOL Grill et al. (2020)	R50 (x2)	93M	4096	62.2	73.5	84.1	91.7
	R50 (x4)	375M	8192	69.1	75.7	87.9	92.5
	R200 (x2)	250M	4096	71.2	77.7	89.5	93.7
VICReg (ours)	R50 (x2)	93M	4096	62.6	73.9	84.5	91.8
	R200 (x2)	250M	4096	68.8	77.3	88.2	93.6

Table 11: **K-NN classifiers on ImageNet.** Top-1 accuracy with 20 and 200 nearest neighbors.

Method	20-NN	200-NN
NPID Wu et al. (2018)	-	46.5
LA Zhuang et al. (2019)	-	49.4
PCL Li et al. (2021)	54.5	-
BYOL Grill et al. (2020)	66.7	64.9
SwAV Caron et al. (2020)	65.7	62.7
Barlow Twins Zbontar et al. (2021)	64.8	62.9
VICReg	64.5	62.8

表9：大型结构线性分类。比较不同方法与编码器结构的 top-1 准确率。VICReg 扩展器输出均为8192维。N-R 为窄型 ResNet，仅加宽瓶颈卷积层。

方法	结构	参数	表征	Top-1	Top-5
SimCLR Chen et al. (2020a)	R50 (x2)	93M	4096	74.2	92.0
	R50 (x4)	375M	8192	76.5	93.2
SwAV Caron et al. (2020)	R50 (x2)	93M	4096	77.3	-
	R50 (x4)	375M	8192	77.9	-
	R50 (x5)	586M	10240	78.5	-
BYOL Grill et al. (2020)	R50 (x2)	93M	4096	77.4	93.6
	R50 (x4)	375M	8192	78.6	94.2
	R200 (x2)	250M	4096	79.6	94.8
VICReg（本文）	N-R50 (x2)	66M	2048	74.7	91.9
	N-R50 (x4)	221M	2048	76.0	92.4
	R50 (x2)	93M	4096	75.5	92.1
	R50 (x4)	375M	8192	75.6	92.2
	RNXT101-32-16	191M	2048	76.1	92.3
	R200 (x2)	250M	4096	77.3	93.3

表10：大型结构半监督分类。比较不同方法与编码器结构的 top-1 准确率。VICReg 扩展器输出均为8192维。

方法	结构	参数	表征	Top-1		Top-5	
				1%	10%	1%	10 %
SimCLR Chen et al. (2020a)	R50 (x2)	93M	4096	58.5	71.7	83.0	91.2
	R50 (x4)	375M	8192	63.0	74.4	85.8	92.6
BYOL Grill et al. (2020)	R50 (x2)	93M	4096	62.2	73.5	84.1	91.7
	R50 (x4)	375M	8192	69.1	75.7	87.9	92.5
	R200 (x2)	250M	4096	71.2	77.7	89.5	93.7
VICReg（本文）	R50 (x2)	93M	4096	62.6	73.9	84.5	91.8
	R200 (x2)	250M	4096	68.8	77.3	88.2	93.6

表11：ImageNet 的 K 近邻分类器。20与200近邻的 top-1 准确率。

方法	20-NN	200-NN
NPID Wu et al. (2018)	-	46.5
LA Zhuang et al. (2019)	-	49.4
PCL Li et al. (2021)	54.5	-
BYOL Grill et al. (2020)	66.7	64.9
SwAV Caron et al. (2020)	65.7	62.7
Barlow Twins Zbontar et al. (2021)	64.8	62.9
VICReg	64.5	62.8

Table 12: **Impact of expander dimensionality.** Top-1 accuracy on the linear evaluation protocol with 100 pretraining epochs.

Dimensionality	256	512	1024	2048	4096	8192	16834
Top-1	55.9	59.2	62.4	65.1	67.3	68.6	68.8

Table 13: **Impact of batch size.** Top-1 accuracy on the linear evaluation protocol with 100 pretraining epochs.

Batch size	128	256	512	1024	2048	4096
Top-1	67.3	67.9	68.2	68.3	68.6	67.8

D.6 Expander network architecture

VICReg borrows the decorrelation mechanism of Barlow Twins Zbontar et al. (2021) and we observe that it therefore has the same dependency on the dimensionality of the expander network. Table 12 reports the impact of the width and depth of the expander network. The dimensionality corresponds the number of hidden and output units in the expander network during pretraining. As the dimensionality increases, the performance dramatically increases from 55.9% top-1 accuracy on linear evaluation with a dimensionality of 256, to 68.8% with dimensionality 16384. The performance tends to saturate as the difference between dimensionality 8192 and 16384 is only of 0.2%.

D.7 Batch size

Contrastive methods suffer from the need of a lot of negative examples which can translate into the need for very large batch sizes Chen et al. (2020a). Table 13 reports the performance on linear classification when the size of the batch varies between 128 and 4096. For each value of batch size, we perform a grid search on the base learning rate described in Appendix C.4. We observe a 0.7% and 1.2% drop in accuracy with small batch size of 256 and 128 which is comparable with the robustness to batch size of Barlow Twins Zbontar et al. (2021) and SimSiam Chen & He (2020), and a 0.8% drop with a batch size of 4096, which is reasonable and allows our method to be very easily parallelized on multiple GPUs.

D.8 Combination with BYOL and SimSiam

BYOL Grill et al. (2020) and SimSiam Chen & He (2020) rely on a effective but difficult to interpret mechanism for preventing collapse, which may lead to instabilities during the training. We incorporate our variance regularization loss into BYOL and SimSiam and show that it helps stabilize the training and offers a small performance improvement. For both methods, the results are obtained using our own implementation and the exact same data augmentation and optimization settings as in their original paper. The variance and covariance regularization losses are incorporated with a factor of $\mu = 1$ for variance and $\nu = 0.01$ for covariance. We report in Figure 3 the improvement obtained over these methods on the linear evaluation protocol for different number of pre-training epochs. For BYOL the improvement is of 0.9% with 100 epochs and becomes less significant as the number of pre-training epochs increases with a 0.2% improvement with 1000 epochs. This indicates that variance regularization makes BYOL converge faster. In SimSiam the improvement is not as significant. We plot in Figure 4 the evolution of the standard deviation computed along each dimension and averaged across the dimensions of the representation and the embeddings, during BYOL and SimSiam pretraining. For both methods, the standard deviation computed on the embeddings perfectly matches $1/\sqrt{d}$ where d is the dimension of the embeddings, which indicates that the embeddings are perfectly spread-out across the unit sphere. This translates in an increased standard deviation at the representation level, which seems to be correlated to the performance improvement. We finally study in Figure 5 the evolution of the average correlation coefficient, during pretraining of BYOL and SimSiam, with and without variance and covariance regularization. The average correlation coefficient is computed by averaging the off-diagonal coefficients of the

表12：扩展器维度的影响。预训练100轮的线性 top-1 准确率。

维度	256	512	1024	2048	4096	8192	16834
Top-1	55.9	59.2	62.4	65.1	67.3	68.6	68.8

表13：批量大小的影响。预训练100轮的线性 top-1 准确率。

批量大小	128	256	512	1024	2048	4096
Top-1	67.3	67.9	68.2	68.3	68.6	67.8

D.6. 扩展网络结构

VICReg 借鉴 Barlow Twins（Zbontar et al., 2021）的去相关机制，因此也依赖扩展网络维度。表12报告扩展器宽度和深度的影响，维度指预训练时隐藏及输出单元数。维度从256增加到16384，线性 top-1 准确率从55.9%显著提高到68.8%；8192与16384仅差0.2个百分点，表现趋于饱和。

D.7. 批量大小

对比方法需要大量负样本，可能因此需要非常大的批量（Chen et al., 2020a）。表13报告批量128至4096的线性分类性能。每种批量均对附录C.4的基础学习率进行网格搜索。批量256、128时准确率分别下降0.7、1.2个百分点，与 Barlow Twins（Zbontar et al., 2021）及 SimSiam（Chen & He, 2020）的批量鲁棒性相当。批量4096时下降0.8个百分点，仍合理，使本方法很容易在多 GPU 上并行。

D.8. 与 BYOL 和 SimSiam 结合

BYOL（Grill et al., 2020）和 SimSiam（Chen & He, 2020）使用有效但难解释的防坍塌机制，可能导致训练不稳定。我们将方差正则损失加入两者，证明它能稳定训练并小幅改善性能。两者均使用我们的实现，数据增强和优化设置完全遵循原论文。方差与协方差正则项系数分别为 $\mu = 1$ 和 $\nu = 0.01$。图3报告不同预训练轮数下线性评估的提升。BYOL 训练100轮时提高0.9个百分点，随着轮数增加，改善减少；1000轮时提高0.2个百分点，说明方差正则化让 BYOL 更快收敛。SimSiam 的改善不太显著。图4展示预训练过程中，沿各维度计算并在维度间平均的标准差，分别针对表征和嵌入。两种方法的嵌入标准差都完全符合 $1/\sqrt{d}$，其中 d 为嵌入维度，说明嵌入充分分散在单位球面上。这进一步提高表征层面的标准差，似乎与性能改善有关。最后，图5研究有无方差与协方差正则化时，BYOL、SimSiam 预训练的平均相关系数变化。平均相关系数通过平均以下矩阵的非对角系数得到：

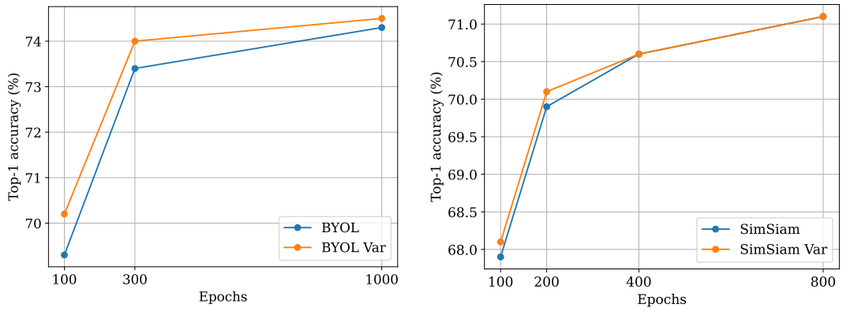


Figure 3: **Incorporating variance regularization in BYOL and SimSiam.** Top-1 accuracy on the linear evaluation protocol for different number of pretraining epochs. For both methods pre-training follows the optimization and data augmentation protocol of their original paper but is based on our implementation. *Var* indicates variance regularization

correlation matrix of the representations:

$$\frac{1}{2d(d-1)} \sum_{i \neq j} C(Y)_{i,j}^2 + C(Y')_{i,j}^2, \tag{9}$$

where Y and Y' are the standardized representations and C is defined in Eq. (3). In BYOL this coefficient is much lower using covariance regularization, which translate in a small improvement of the performance, according to Table 4. We do not observe the same improvement in SimSiam, both in terms of correlation coefficient, and in terms of performance on linear classification. The average correlation coefficient is correlated with the performance, which motivates the fact that decorrelation and redundancy reduction are core mechanisms for learning self-supervised representations.

E RUNNING TIME

We report in Table 14, the running time of VICReg in comparison with other methods. All methods are run by us on 32 Tesla V100 GPUs. Each method offers a different trade-off between running time, memory and performance. SwAV is a very fast algorithm which use less memory and run faster than the other methods but with a lower performance, multi-crop helps the performance at the cost of additional compute and memory usage. BYOL has the highest memory requirement, which is due to the need of storing the target network weights. Finally, Barlow Twins and VICReg offer an interesting trade-off, consuming less memory than BYOL and SwAV with multi-crop, and running faster than SwAV with multi-crop, but with a slightly worse performance. The difference of 1h running time between Barlow Twins and VICReg is probably due to implementation details not related to the method.

Table 14: **Running time and peak memory.** Comparison between different methods, the training is distributed on 32 Tesla V100 GPUs, the running time is measured over 100 epochs and the peak memory is measured on a single GPU. We report top-1 accuracy (%) on linear classification on top of the frozen representations.

Method	time / 100 epochs	peak memory / GPU	Top-1 accuracy (%)
SwAV	9h	9.5G	71.8
SwAV (w/ multi-crop)	13h	12.9G	75.3
BYOL	10h	14.6G	74.3
Barlow Twins	12h	11.3G	73.2
VICReg	11h	11.3G	73.2

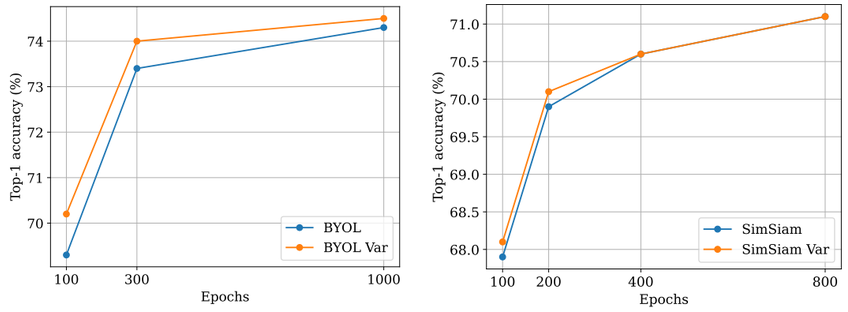


图3：在 BYOL 和 SimSiam 中加入方差正则。不同预训练轮数的线性 top-1 准确率。两者使用我们的实现，优化和增强遵循原论文。Var 表示方差正则。

表征的相关矩阵：

$$\frac{1}{2d(d-1)} \sum_{i \neq j} C(Y)_{i,j}^2 + C(Y')_{i,j}^2, \tag{9}$$

其中 Y、Y' 是标准化表征，C 定义见式3。BYOL 加协方差正则后，此系数明显更低，并如表4所示带来小幅性能提升。SimSiam 的相关系数和线性分类性能均未出现同样改善。平均相关系数与性能相关，支持去相关和冗余消减是自监督表征学习的核心机制。

E. 运行时间

表14比较 VICReg 与其他方法的运行时间，全部由我们在32块 Tesla V100 上运行。各方法在时间、内存与性能间有不同权衡。SwAV 速度快、内存少，但性能较低；多裁剪以额外计算和内存换取提升。BYOL 因需存储目标网络权重而内存需求最高。Barlow Twins 与 VICReg 的权衡较有吸引力：内存少于 BYOL 和多裁剪 SwAV，速度快于多裁剪 SwAV，但性能略低。Barlow Twins 与 VICReg 相差1小时，可能源于与方法无关的实现细节。

表14：运行时间与峰值内存。32块 Tesla V100 分布式训练，时间按100轮测量，峰值内存按单 GPU 测量。报告冻结表征线性分类 top-1 准确率（%）。

方法	时间／100轮	峰值内存／GPU	Top-1 准确率（%）
SwAV	9h	9.5G	71.8
SwAV（多裁剪）	13h	12.9G	75.3
BYOL	10h	14.6G	74.3
Barlow Twins	12h	11.3G	73.2
VICReg	11h	11.3G	73.2

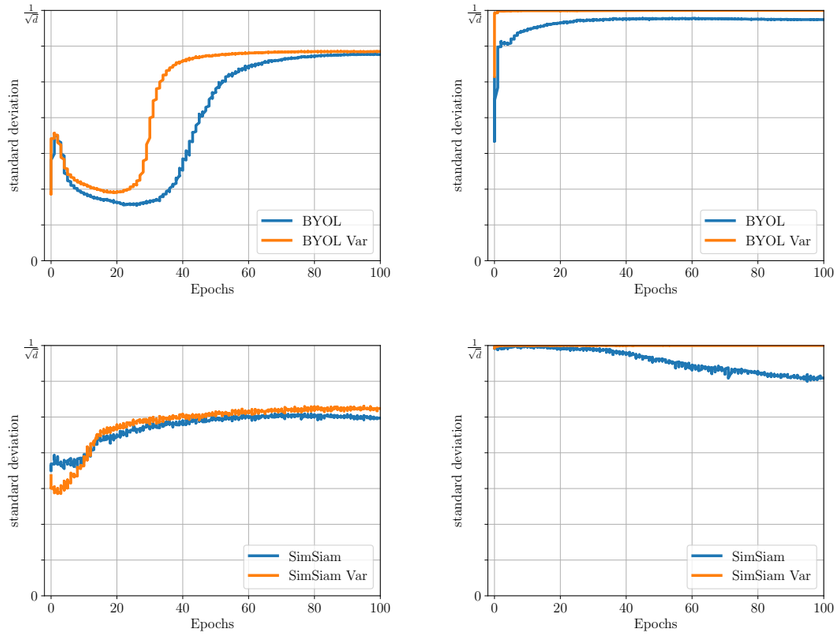


Figure 4: **Standard deviation of the features during BYOL and SimSiam pretraining.** Evolution of the average standard deviation of each dimension of the features with and without variance regularization (Var). left: the standard deviation is measured on the representations, right: the standard deviation is measured on the embeddings.

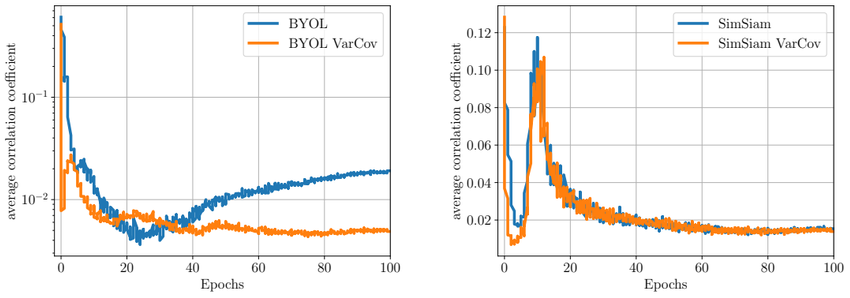


Figure 5: **Average correlation coefficient of the features during BYOL and SimSiam pretraining.** Evolution of the average correlation coefficient measured by averaging the off-diagonal terms of the correlation matrix of the representations with BYOL, BYOL with variance-covariance regularization (BYOL VarCov), SimSiam, and SimSiam with variance-covariance regularization (SimSiam VarCov).

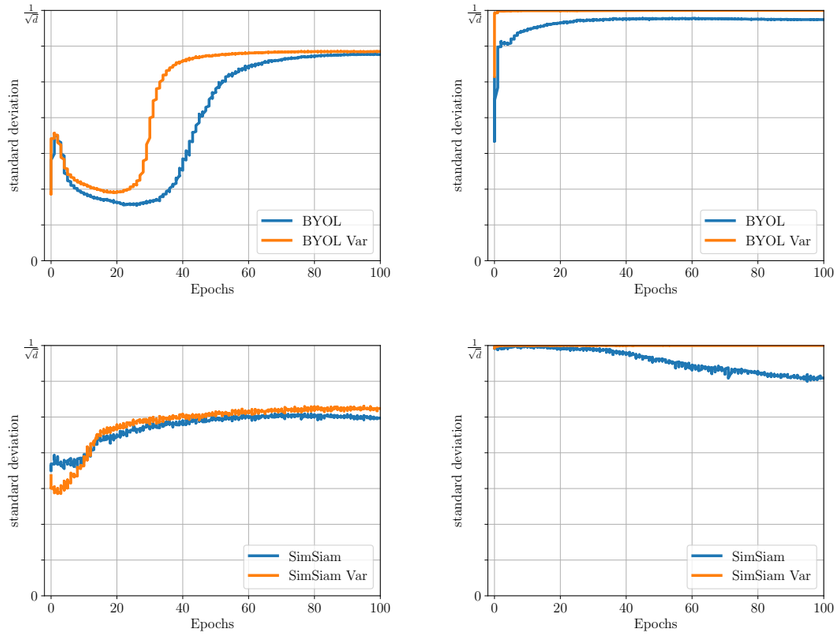


图4：BYOL 与 SimSiam 预训练特征标准差。有无方差正则（Var）时，各特征维度标准差的平均值变化。左：表征；右：嵌入。

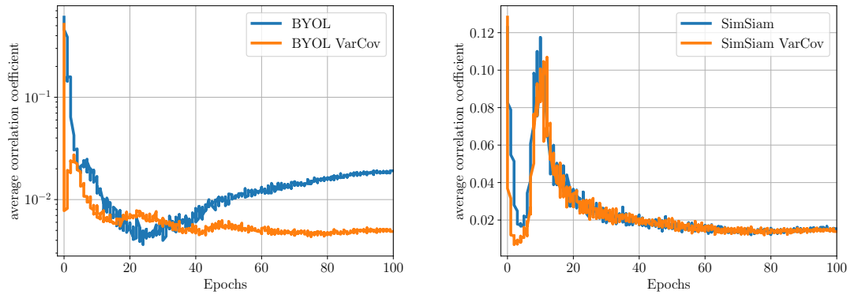


图5：BYOL 和 SimSiam 预训练特征的平均相关系数。通过平均表征相关矩阵非对角项，比较 BYOL、加方差—协方差正则的 BYOL（BYOL VarCov）、SimSiam、加方差—协方差正则的 SimSiam（SimSiam VarCov）的变化。